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Macao Polytechnic University

Class 7: Healthy Ageing

Geriatrics

Kefeng Li, PhD

E-mail: kefengl@mpu.edu.mo

Office: New Academic Building E1303-1 (Main Campus, Macau);

Bldg 11 Room 206 (Hengqin Campus)

Course webpage: https://canvas.mpu.edu.mo/

Office hours : Wednesday: 10 am -1 pm, and Thursday: 10 am - 1 pm

Healthy Ageing

- Aging is a natural process, and we will all get older
- Our philosophy is simple:

  what we truly want is

  Healthy Aging

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Care Knowledge Primer for Smart Healthy Ageing

Introduction | Frontiers in Smart Healthy Ageing

The goal is not only to extend life, but to preserve function, independence, and wellbeing.

- **Ageing mechanisms**
- **Functional assessment**
- **Care data streams**
- **AI-ready questions**

Healthspan, Not Just Lifespan, Is the Core Target

Ageing research becomes care research when extra years are translated into functional ability.

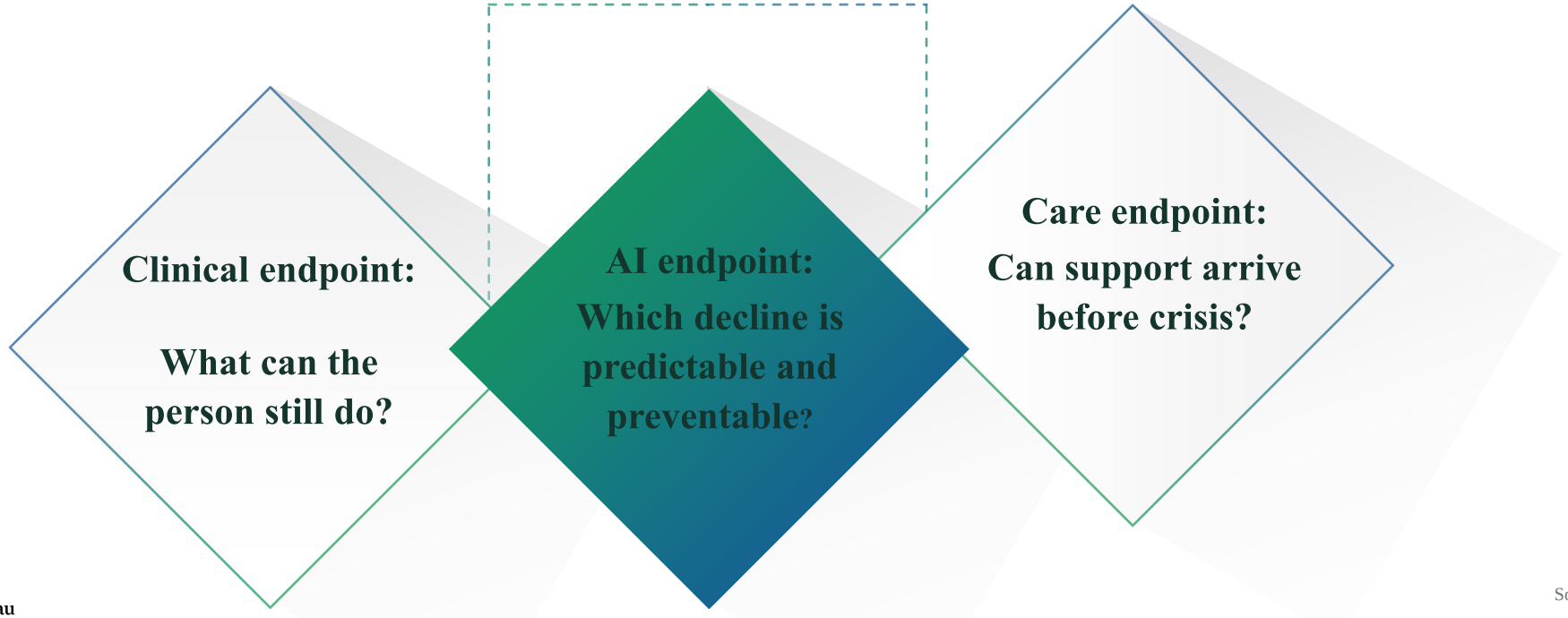
Birth

Later life

Lifespan

Healthspan: preserved capacity | **Disability gap**

Healthy ageing reframes success from survival alone to wellbeing through intrinsic capacity, environment, and participation.



Sources: WHO; GBD 2021

Smart Ageing Requires a Shared Cross-Disciplinary Lens

Geriatrics
syndromes + function

Public health
prevention + equity

Same problem, different variables

Same person, different interventions

AI / data science
prediction + explanation

Engineering
Sensing + robotics

Ethics and social care
privacy + dignity

Sources: WHO ICOPE; course syllabus

Learning Map: From Basics to Research Questions

Why ageing matters
healthspan, function, quality of life

1

2

Mechanisms
cellular and systemic drivers

Geriatric syndromes
frailty, falls, delirium, polypharmacy

3

4

Assessment
ADL, IADL, mobility, cognition, QoL

Data bridge
EHR, wearables, imaging, voice, behavior

5

Foundation for later modules: models are only meaningful when their labels and targets reflect real care needs.

Instrumental Activities of Daily Living (IADLs); Activities of Daily Living (ADLs)

Sources: WHO healthy ageing; Hallmarks of Aging

A Single Case Shows Why Elderly Care is Complicated

Fictional vignette: Ms. Li, 79, lives alone

Living alone:

low social support, delayed help-seeking

Fall risk:

Slow gait + cane + night waking

Memory concern:

Missed appointments and repeated questions

Polypharmacy:

Multiple medicines, uncertain adherence

Insomnia:

Fragmented sleep and daytime fatigue

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Sources: Inouye et al.; WHO ICOPE; Falls guidelines

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Basic Concepts in Smart Healthy Ageing

Ageing, healthy ageing, care, and smart care systems

Before building AI models, we need a shared language for what ageing care is trying to preserve

- Ageing
- Healthy ageing
- Care
- Smart care

The Same Word Family Describes Different Levels of Work

	Goal
Ageing — Demographic and biological change over time	**What is changing?**
Healthy ageing — Developing and maintaining functional ability	**What should be preserved?**
healthy care — Wellness-oriented support across health, living, and social participation	**What support ecosystem is needed?**
Care — Assessment, prevention, treatment, rehabilitation, and daily support	**What actions are delivered?**
Smart healthy ageing — Data-enabled care that senses, predicts, explains, and assists	**How can technology improve timing and personalization?**

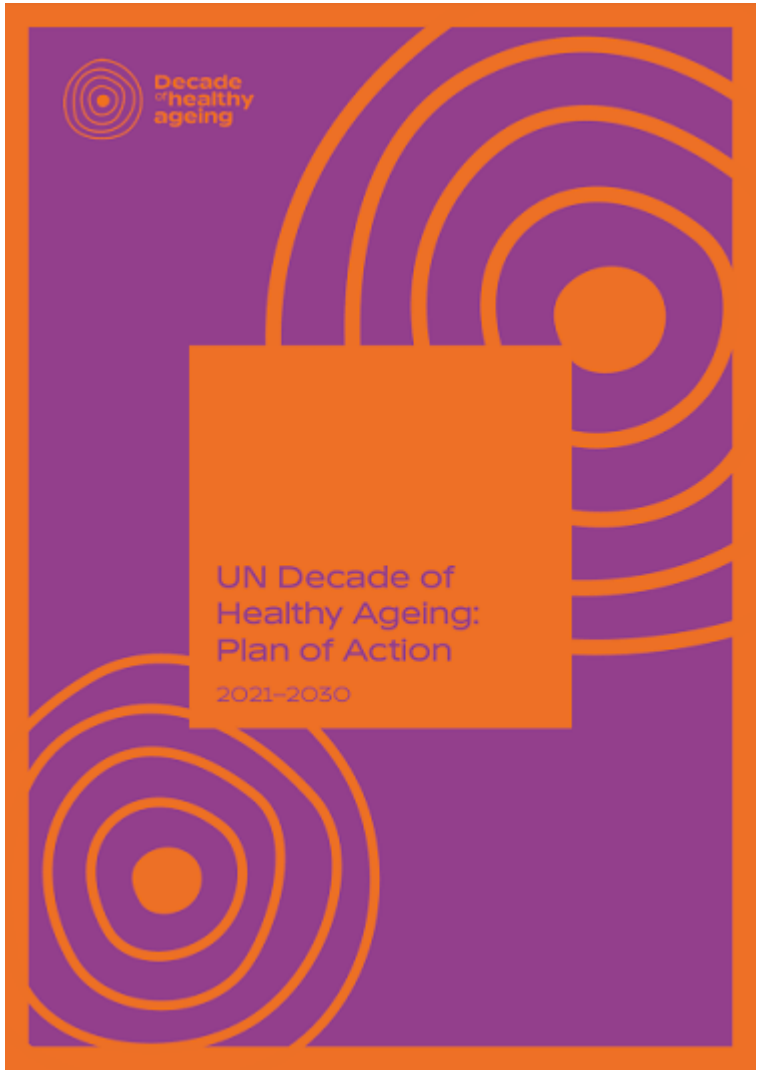
WHO healthy ageing; UN WPA 2023

WHO Healthy Ageing Framework (2030)

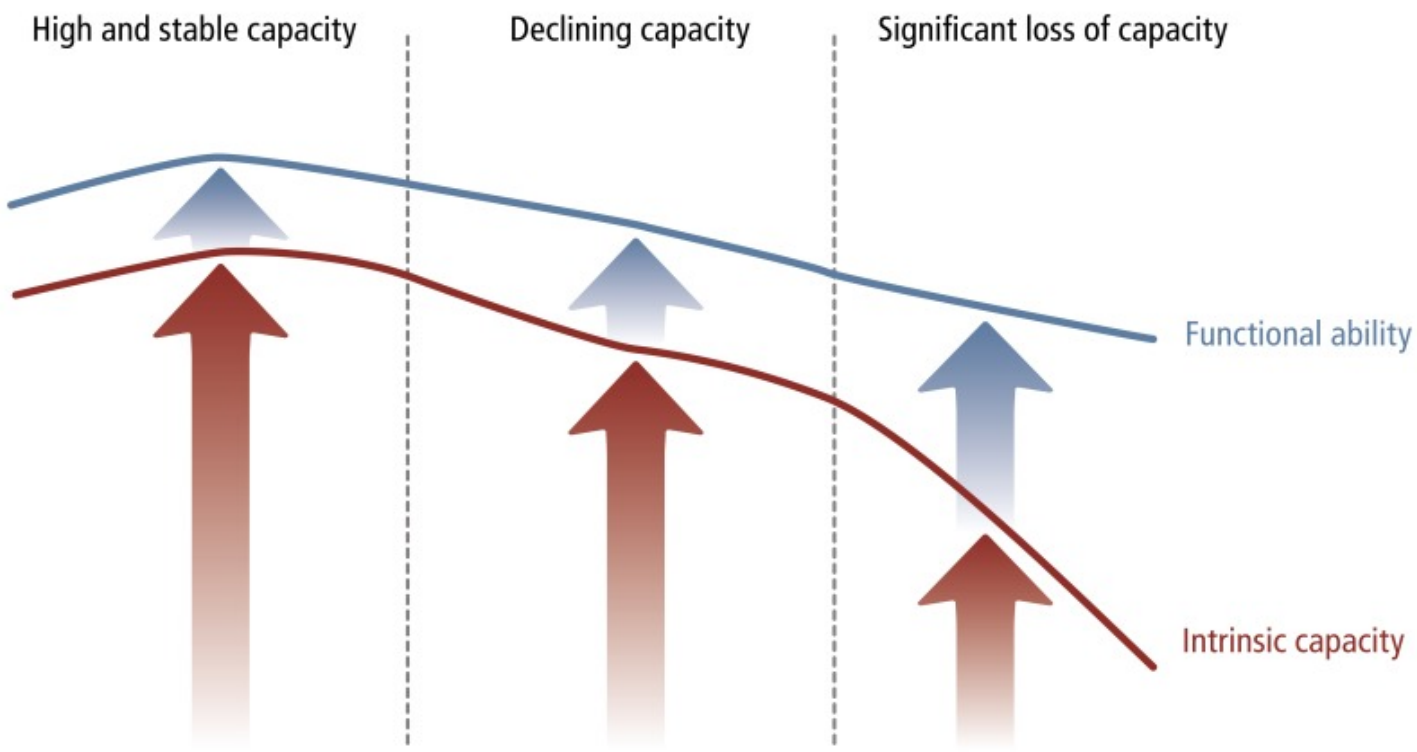
- **Every person – in every country in the world – should have the opportunity to live a long and healthy life.**

- WHO defines healthy ageing as "**the process of developing and maintaining the functional ability that enables wellbeing in older age.**"

- Functional ability is about having the capabilities that enable all people to be and do what they have reason to value.
    - Meet their basic needs;
    - Learn, grow and make decisions;
    - Be mobile;
    - Build and maintain relationships; and
    - Contribute to society.



Healthy Ageing is About Functional Ability in Context



- **Intrinsic capacity**

The composite of physical and mental capacities a person can draw on.

- **Functional ability**

What a person can actually do in a real environment.

- **Environment can compensate:**

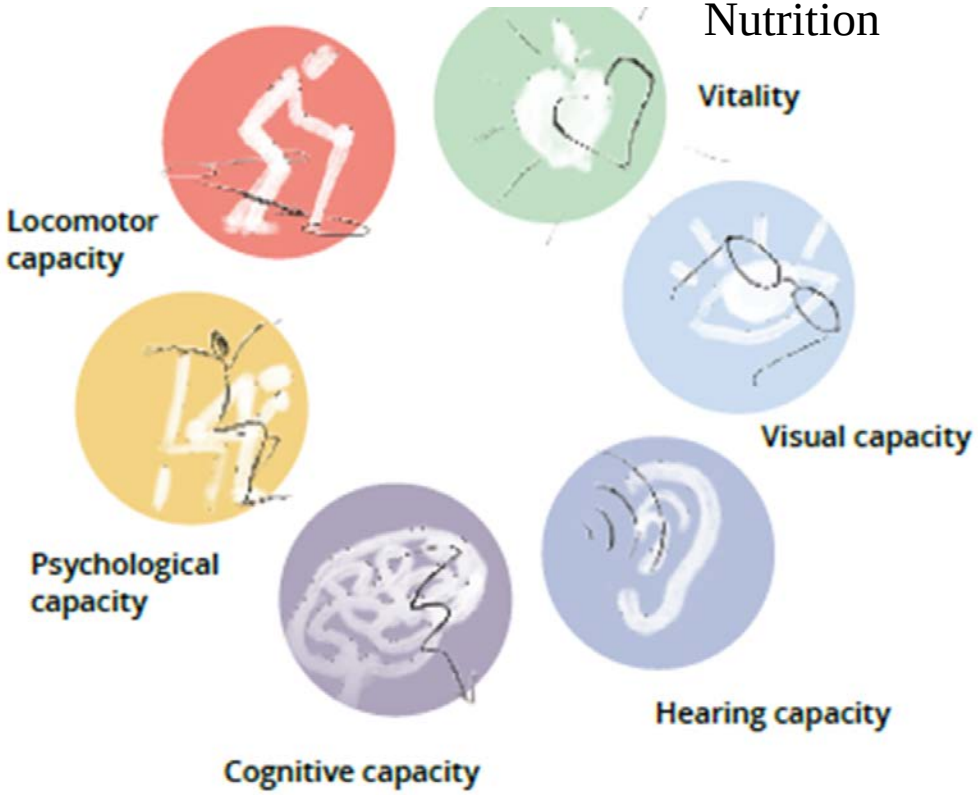
Assistive devices, housing, caregivers, services, and policy can keep ability higher than capacity alone.

This is the conceptual bridge from biology to care design.

WHO World report, Fig. 1.1

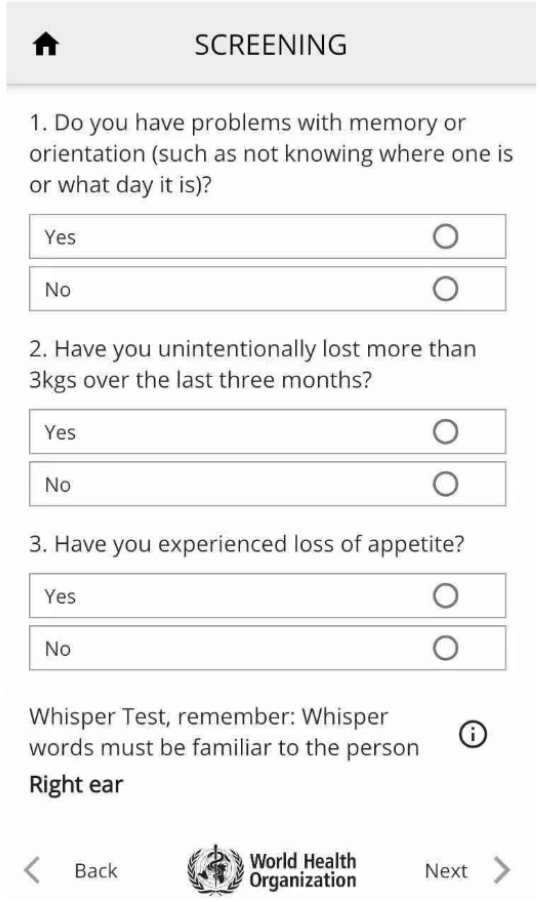
How do we evaluate intrinsic capacity and functional ability?

Intrinsic capacity (IC)



Six key domains of IC

The Integrated Care for Older People (ICOPE)



A digital tool developed by the WHO to help health and social care workers implement person-centred, coordinated care for older adults

How do we evaluate intrinsic capacity and functional ability?

Leung et al. BMC Geriatrics (2022) 22:304
https://doi.org/10.1186/s12877-022-02980-1

BMC Geriatrics

RESEARCH Open Access

Intrinsic capacity of o ople in the community usi D Integrated Care for Older People (ICOPE) framework: a cross-sectional study

已添加到剪贴板

Angela Y. M. Leung[1,2,3], Jing Jing Su[1,2,3], Elsa S. H. Lee[4], Jeff T. S. Fung[3] and Alex Molassiotis[1,3*]

How do we evaluate intrinsic capacity and functional ability?

Intrinsic capacity (IC) Initial Screening

Priority conditions associated with declines in intrinsic capacity	Tests	Assess fully any domain with a checked circle
COGNITIVE DECLINE (Chapter 4)	1. Remember three words: flower, door, rice (for example)	
	2. Orientation in time and space: What is the full date today? Where are you now (home, clinic, etc)?	○ Wrong to either question or does not know
	3. Recalls the three words?	○ Cannot recall all three words
LIMITED MOBILITY (Chapter 5)	Chair rise test: Rise from chair five times without using arms. Did the person complete five chair rises within 14 seconds?	○ No
MALNUTRITION (Chapter 6)	1. Weight loss: Have you unintentionally lost more than 3 kg over the last three months?	○ Yes
	2. Appetite loss: Have you experienced loss of appetite?	○ Yes
VISUAL IMPAIRMENT (Chapter 7)	Do you have any problems with your eyes: difficulties in seeing far, reading, eye diseases or currently under medical treatment (e.g. diabetes, high blood pressure)?	○ Yes
HEARING LOSS (Chapter 8)	Hears whispers (whisper test) *or*	
	Screening audiometry result is 35 dB or less *or*	○ Fail
	Passes automated app-based digits-in-noise test	
DEPRESSIVE SYMPTOMS (Chapter 9)	Over the past two weeks, have you been bothered by	
	– feeling down, depressed or hopeless?	○ Yes
	– little interest or pleasure in doing things?	○ Yes

How do we evaluate intrinsic capacity and functional ability?

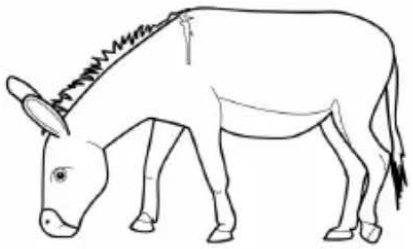
ICOPE for Intrinsic Capacity (IC) Assessment

Cognitive	Montreal cognitive assessment (MoCA)
Vitality	Mini Nutritional Assessment (MNA)
Psychological	Patient Health Questionnaire (PHQ-9)
Hearing	Whisper test; Webber & Rinne
Visual	Visual acuity

The Montreal Cognitive Assessment (MoCA)

MoCA is a quick 10-minute screening tool used by health care workers to check for mild cognitive impairment and early dementia

NAMING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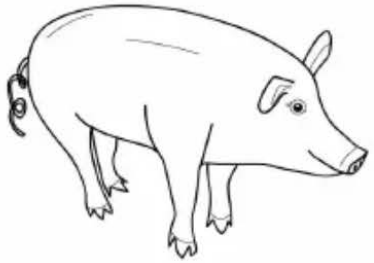
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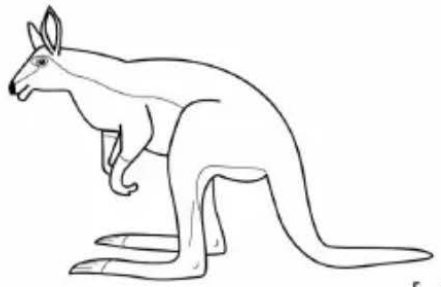

[ ] __/3

MEMORY Read list of words, subject must repeat them. Do 2 trials, even if 1st trial is successful. Do a recall after 5 minutes.

	TRAIN	EGG	HAT	CHAIR	BLUE
1st trial					
2nd trial					

No points

ATTENTION Read list of digits (1 digit/ sec.).

Subject has to repeat them in the forward order [ ] 5 4 1 8 7

Subject has to repeat them in the backward order [ ] 1 7 4 __/2

Read list of letters. The subject must tap with his hand at each letter A. No points if ≥ 2 errors

[ ] F B A C M N A A J K L B A F A K D E A A A J A M O F A A B __/1

Serial 7 subtraction starting at 80 [ ] 73 [ ] 66 [ ] 59 [ ] 52 [ ] 45

4 or 5 correct subtractions: **3 pts**, 2 or 3 correct: **2 pts**, 1 correct: **1 pt**, 0 correct: **0 pt** __/3

LANGUAGE Repeat : She heard his lawyer was the one to sue after the accident. [ ]

The little girls who were given too much candy got stomach aches. [ ] __/2

Fluency / Name maximum number of words in one minute that begin with the letter B [ ] _____ (N ≥ 11 words) __/1

ABSTRACTION Similarity between e.g. banana - orange = fruit [ ] eye – ear [ ] trumpet – piano __/2

How do we evaluate intrinsic capacity and functional ability?

ICOPE for Intrinsic Capacity (IC) Assessment

Locomotor	Short Physical Performance Battery (SPPB) A hierarchical test of **standing balance, a 4-meter walk test, and five repetitive chair stands** Grip strength; Mobile phone physical activity level questionnaire
Social care and support	Level of **self-care capacity** in activities of daily living; the housing and financial status; the level of loneliness; and extent of **social engagement**.

How do we evaluate intrinsic capacity and functional ability?

ICOPE for Intrinsic Capacity (IC) Assessment

Short Physical Performance Battery (SPPB)

How do we evaluate intrinsic capacity and functional ability?

Functional Ability

Activities of Daily Living (ADLs) / Barthel Index

Contains 10 items covering basic self-care and mobility tasks.

Activity	Independent	Needs Some Help	Cannot Do
Bathing: Able to bathe independently, including getting in and out of the tub or shower	☐	☐	☐
Dressing: Able to select clothing and dress and undress independently	☐	☐	☐
Toileting: Able to get to the toilet, use it, and manage clothing and hygiene independently	☐	☐	☐
Transferring: Able to move from bed to chair, sit down and stand up without assistance	☐	☐	☐
Continence: Able to control bladder and bowel function	☐	☐	☐
Feeding / Eating: Able to feed themselves once food is prepared	☐	☐	☐

IADLs: Instrumental Activities of Daily Living

(IADL) Scale consists of exactly 8 items. These items assess more complex skills needed for an older adult to live independently in the community.

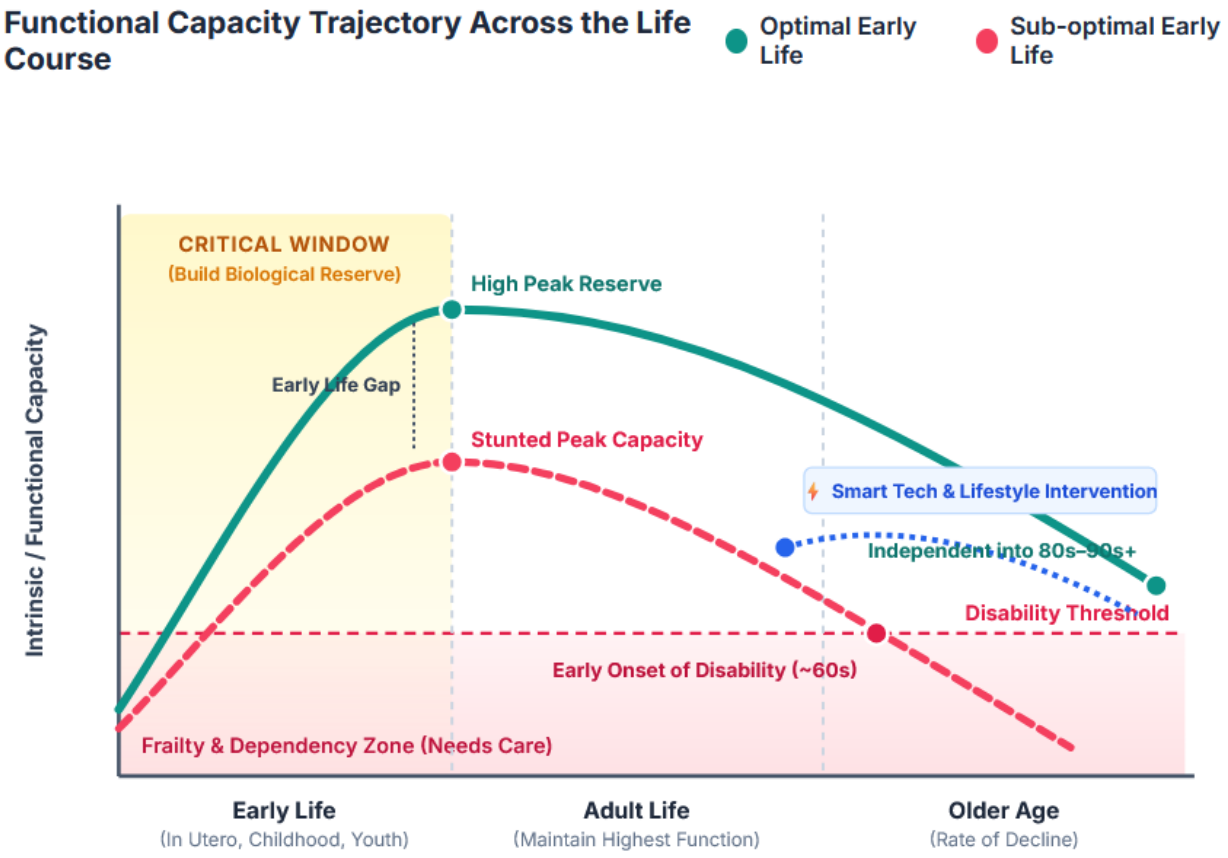
Using the telephone: Able to look up numbers, dial, and use the phone independently	☐	☐	☐	_____
Shopping: Able to plan and complete grocery or personal shopping independently	☐	☐	☐	_____
Food preparation: Able to plan and prepare meals independently	☐	☐	☐	_____
Housekeeping: Able to maintain the home at a safe, livable standard	☐	☐	☐	_____
Laundry: Able to wash, dry, and put away clothing and linens independently	☐	☐	☐	_____
Transportation: Able to drive or arrange independent transportation	☐	☐	☐	_____
Managing medications: Able to manage own prescriptions, refills, and dosing schedule	☐	☐	☐	_____
Managing finances: Able to handle banking, bill payment, and financial decisions	☐	☐	☐	_____

IADL Summary: Independent: ___ / 8 Needs Some Help: ___ / 8 Cannot Do: ___ / 8

A Life Course Approach: Early Life Matters

Quantifying Intrinsic Capacity Across the Lifespan

Ageing begins at conception, not at age 60



Adapted from WHO Life Course Model (Kalache & Kickbusch, 1997). Peak capacity attained in youth dictates functional longevity.

1. DOHaD & Epigenetic Programming

Developmental Origins of Health & Disease (Barker Hypothesis): The *first 1,000 days* (in utero to age 2) program metabolic, cardiovascular, and immune systems via DNA methylation, shaping chronic disease risk 60 years later.

2. Building "Biological & Cognitive Reserve"

Peak bone mass, muscle fiber number, and neural synaptic density are established before age 25. **Higher early-life peaks** act as a buffer—delaying when age-related decline crosses the *disability threshold*.

3. Cumulative Exposure & "Chains of Risk"

Early socioeconomic adversity, malnutrition, or toxic stress accelerates **epigenetic ageing clocks** and inflammaging. Advantage or disadvantage compounds exponentially across decades.

4. Frontiers in Smart Healthy Ageing

Paradigm Shift: Moving from *geriatric reactive care* to **lifelong precision prevention**. Using Digital Twins, AI multi-omics, and early-life digital biomarkers to forecast and bend ageing trajectories early.

Disease Model vs Functional Model

Disease model		Functional model
What disease is present?		What matters in daily life?
Which biomarker is abnormal?	**versus**	Which capacity is declining?
Which treatment reduces pathology?		Which support preserves participation?
Success: disease control		Success: meaningful ability

Same diagnosis, different care plan:
Two patients with diabetes may have very different mobility, cognition, home support, and life goals.

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doi:10.1097/00006199-199407000-00002

Elderly Care is Shifting From Treating Disease to Maintaining Ability

Treat
diagnose
and manage diseases

Prevent
identify risk
before crisis

Rehabilitate
recover or compensate
lost capacity

Support
adapt environments
and routines

Maintain
preserve autonomy,
participation, QoL

Design rule for smart care: predict a change only if it can trigger a meaningful assessment or intervention.

Examples: fall-risk alert -> home safety review | sleep deterioration -> medication review | low activity -> rehab plan

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doi:10.1155/2019/1354362

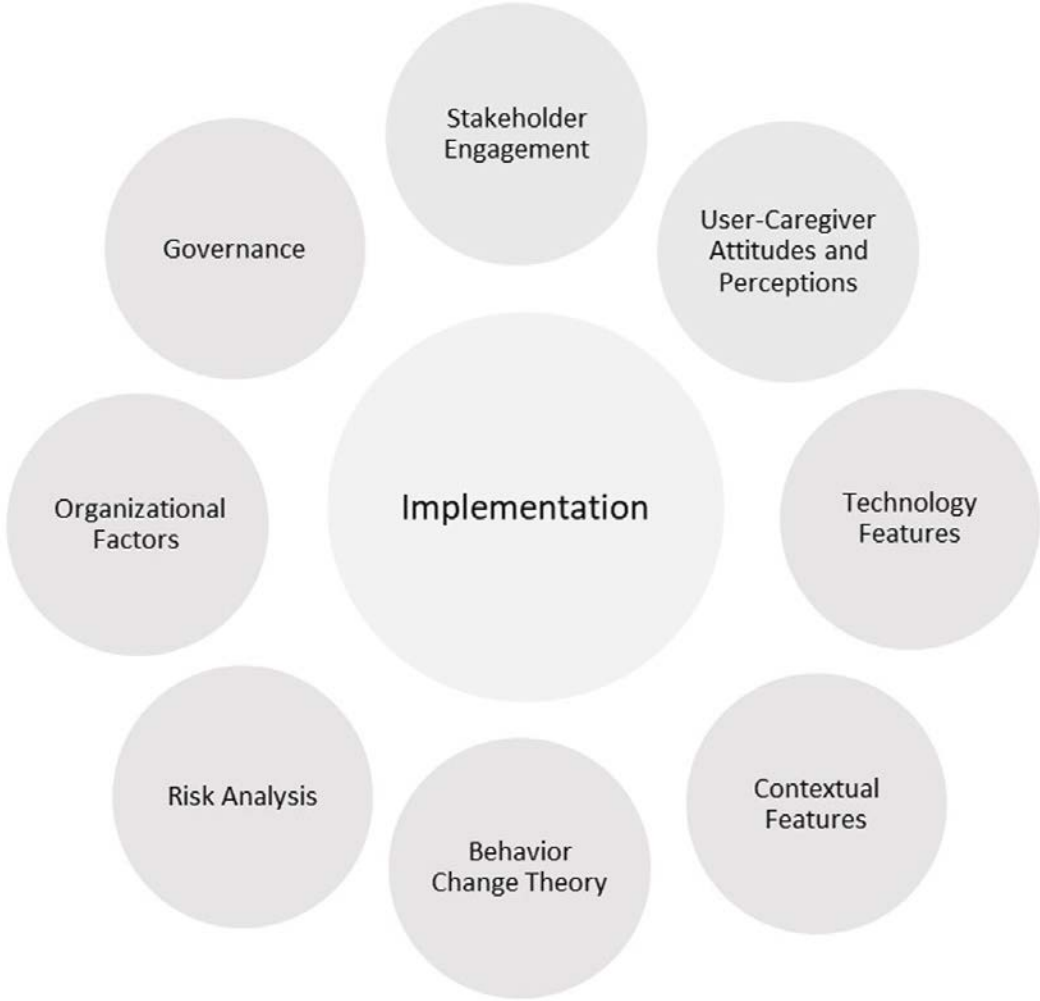
Where Medicine, Engineering, AI, and Public Health Meet

- **Medicine**
  clinical meaning risk, syndromes, outcomes
- **Engineering**
  measurement sensors, devices, robotics

Smart care system

- **AI**
  inference prediction, explanation, agents
- **Public health**
  systems prevention, equity, scale

Technology matters only when it fits people, caregivers, organizations, governance, and risk.



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doi:10.3389/fragi.2024.1349520

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Introduction to Mechanisms of Ageing

Biological ageing, biological age, healthspan, and hallmarks

Ageing is best understood as a multi-level loss of resilience, not as a single disease or a simple clock.

- chronological age
- biological age
- healthspan
- 12 hallmarks

Biological Ageing is a Progressive Loss of Resilience

Biological ageing:

time-dependent molecular and cellular damage that reduces physiological integrity and increases vulnerability.

Molecular stress
DNA damage
protein errors
metabolic stress

>

Cellular response
senescence
autophagy change
mitochondrial stress

>

Tissue dysfunction
inflammation
stem-cell exhaustion
loss of repair

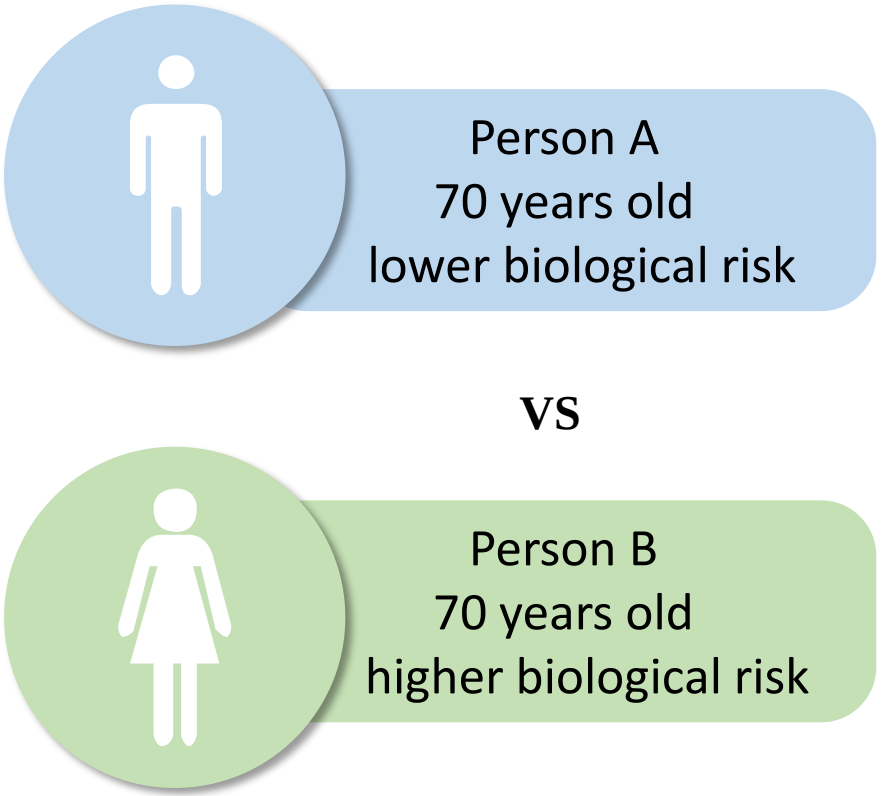
>

Care signal
frailty
functional decline
disease vulnerability

For smart care, mechanisms matter because they explain why small stresses can become visible functional decline.

doi:10.1016/j.cell.2013.05.039; 10.1016/j.cell.2022.11.001

Chronological Age vs Biological Age

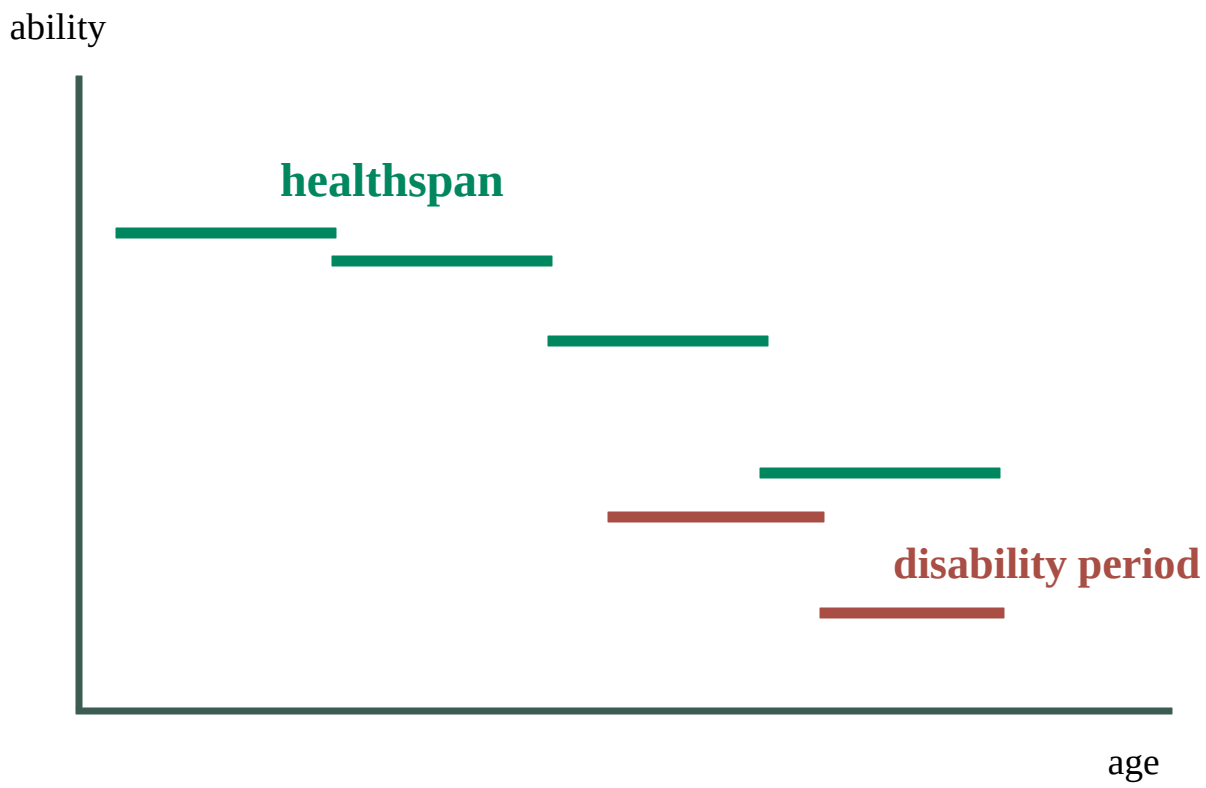


- **Epigenetic clocks**
  DNA methylation patterns
- **Clinical biomarkers**
  inflammation, glucose, kidney, lipids
- **Physiology**
  grip strength, gait speed, VO2
- **Multi-omics**
  transcriptome, proteome, metabolome

same chronological age != same biological state

doi:10.1186/gb-2013-14-10-r115; 10.18632/aging.101414

Healthspan is the Period Before Disability Dominates



Lifespan asks:
How long did the person live?

Healthspan asks:
How long did the person live with capacity and participation?

Smart care aim:
shift decline later, detect risk earlier, and compress disability.

adapted from doi:10.1056/NEJM198007173030304

Ageing is Not Driven by One Cause

Primary hallmarks

initial damage and maintenance failures

Antagonistic hallmarks

compensatory responses that become harmful

Integrative hallmarks

system-level consequences and clinical decline

- **Network logic:**

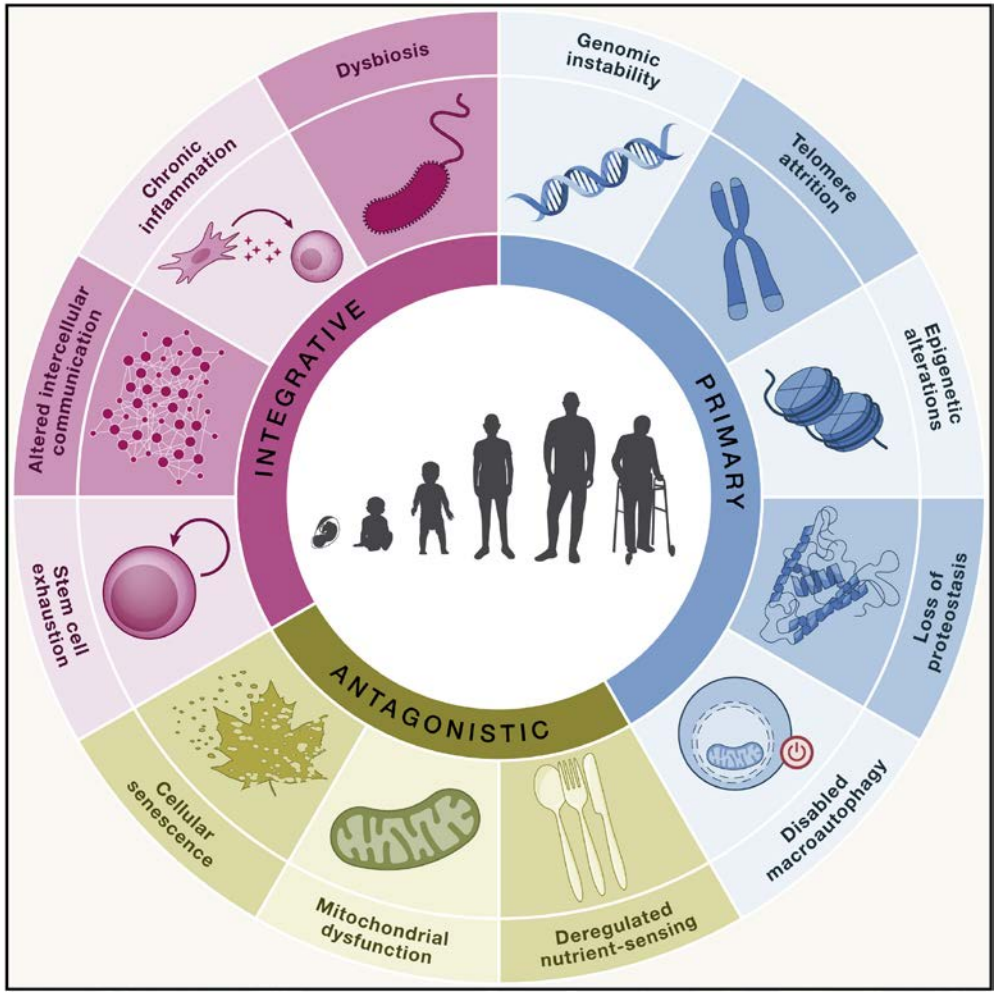
DNA damage, nutrient sensing, mitochondria, inflammation, and microbiome interact across scales.

- **Care implication:**

No single biomarker or device captures ageing; robust assessment combines mechanisms, function, and context.

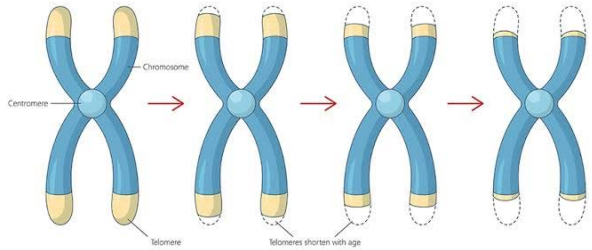
adapted from doi:10.1016/j.cell.2022.11.001

The 12 Hallmarks Provide a Mechanistic Map



- **Primary**
  - Genomic instability
  - Telomere attrition
  - Epigenetic alterations
  - Loss of proteostasis
  - Disabled macroautophagy
- **Antagonistic**
  - Deregulated nutrient-sensing
  - Mitochondrial dysfunction
  - Cellular senescence
- **Integrative**
  - Stem cell exhaustion
  - Altered intercellular communication
  - Chronic inflammation
  - Dysbiosis

Telomere Shortening With Age



Beginner takeaway: the hallmarks are linked mechanisms that turn biological change into functional vulnerability.

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doi:10.1016/j.cell.2022.11.001

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Advanced Mechanisms of Ageing

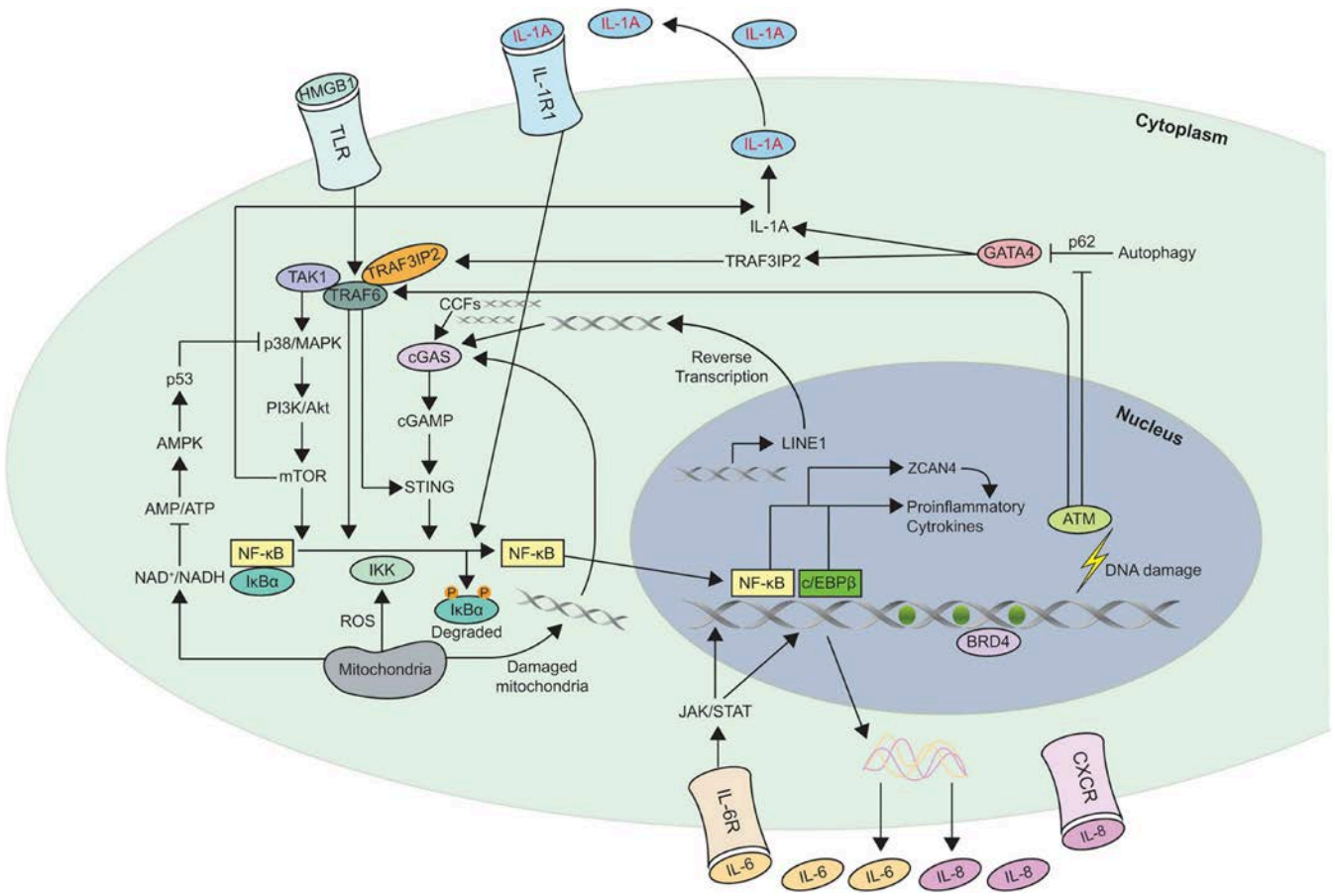
From cellular pathways to organ function and care needs

Advanced mechanisms matter when they help explain why an organ becomes vulnerable, which function is lost, and what care need appears next.

cell signals | system inflammation | energy sensing | organ ageing | care needs

doi:10.1016/j.heliyon.2024.e24751

Senescent Cells (Ageing Cells) Turn Damage into Tissue-Level Signals



SASP regulation connects DNA damage, mitochondria, cGAS-STING and NF-kB.

➢ Basic idea

Cellular senescence is a durable growth-arrest state triggered by stress such as DNA damage, oncogene signaling, telomere attrition or mitochondrial dysfunction.

➢ Pathway logic

p53/p21 and p16/Rb help lock cell-cycle arrest, while NF-kB, C/EBPβ, cGAS-STING and mitochondrial ROS shape the secreted SASP profile.

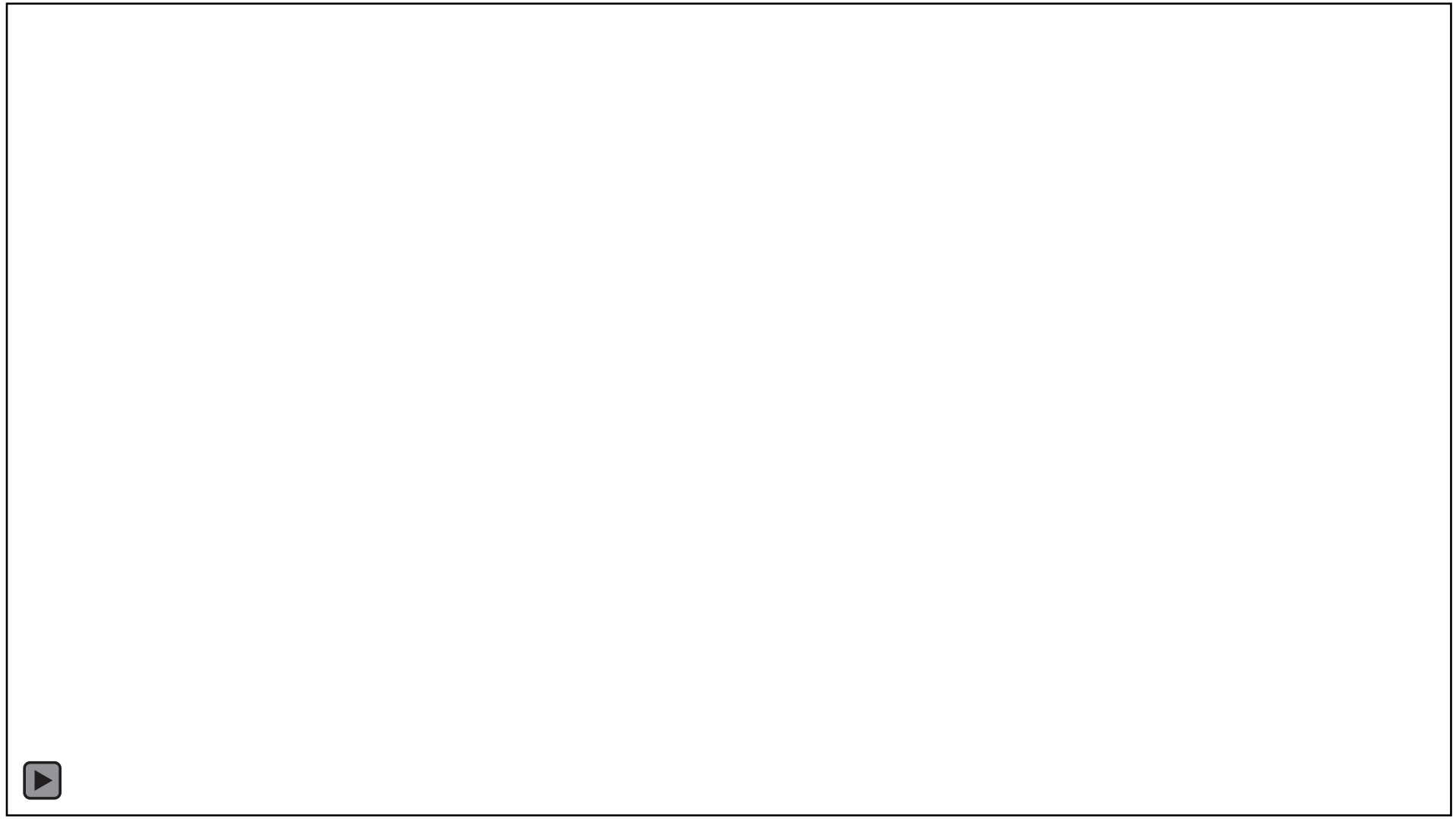
➢ Care relevance

Transient senescence can aid repair; persistent SASP can amplify pain, fibrosis, frailty, poor wound healing and multimorbidity risk.

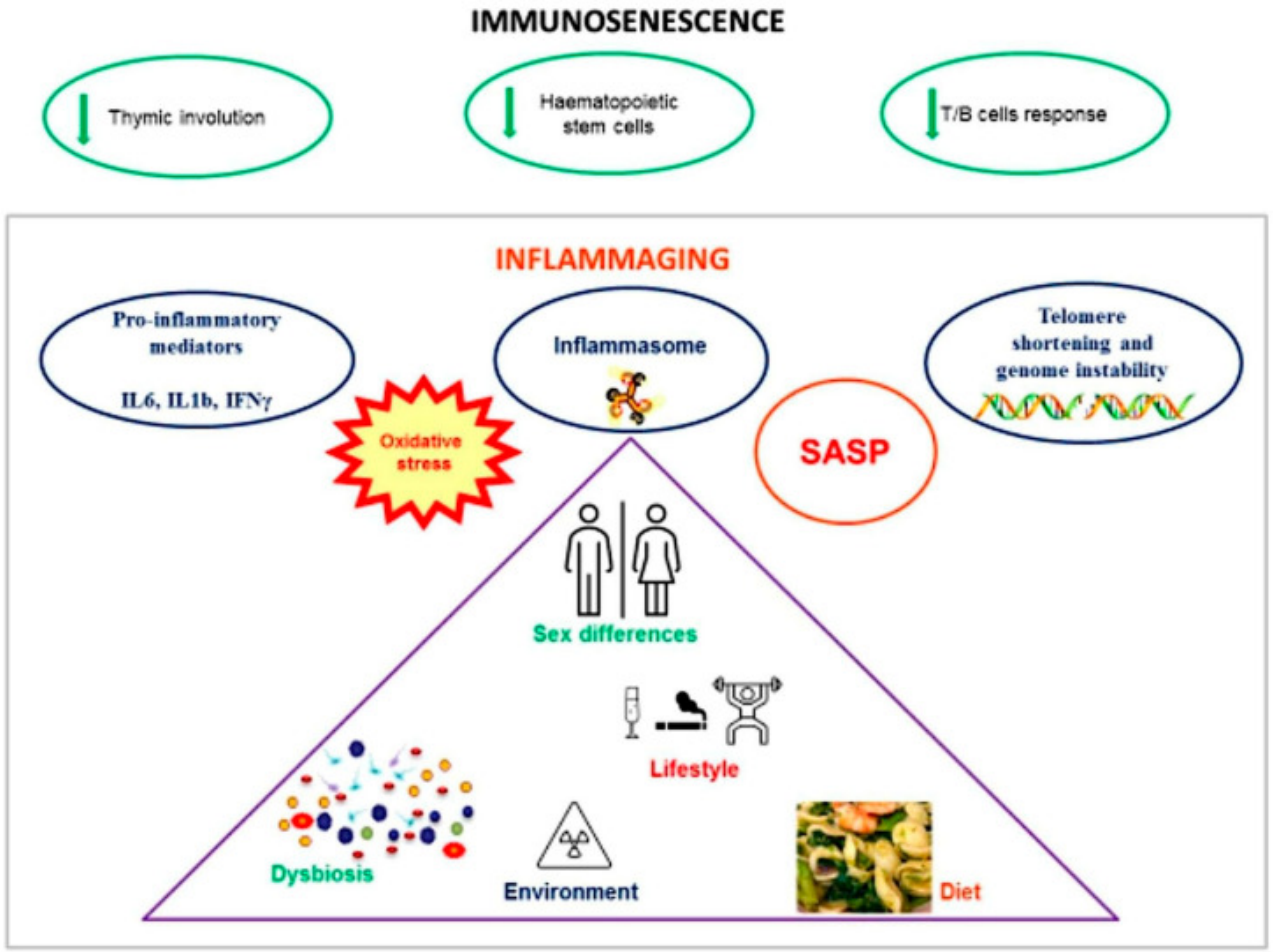
doi:10.3389/fcell.2021.645593

Senescent Cells (Ageing Cells) Turn Damage into Tissue-Level Signals

Senescent cells video



Inflammaging Converts Local Damage into Systemic Risk



Low-grade inflammation is sustained by senescence, dysbiosis, oxidative stress and immune ageing.

Inflammaging is chronic, low-grade, mostly sterile inflammation that rises with age even without acute infection.

- Sources: senescent-cell SASP, damaged macromolecules, mitochondrial ROS, gut barrier leakage and lifelong antigen exposure.
- Typical markers discussed in studies include IL-6, TNF-alpha, CRP, IL-1 family signals and inflammasome activity.
- Clinical meaning: the baseline is already inflamed, so infection, surgery, sleep loss or immobility can push function over a threshold.

Think of inflammaging as a background amplifier: it makes small stressors produce larger functional consequences.

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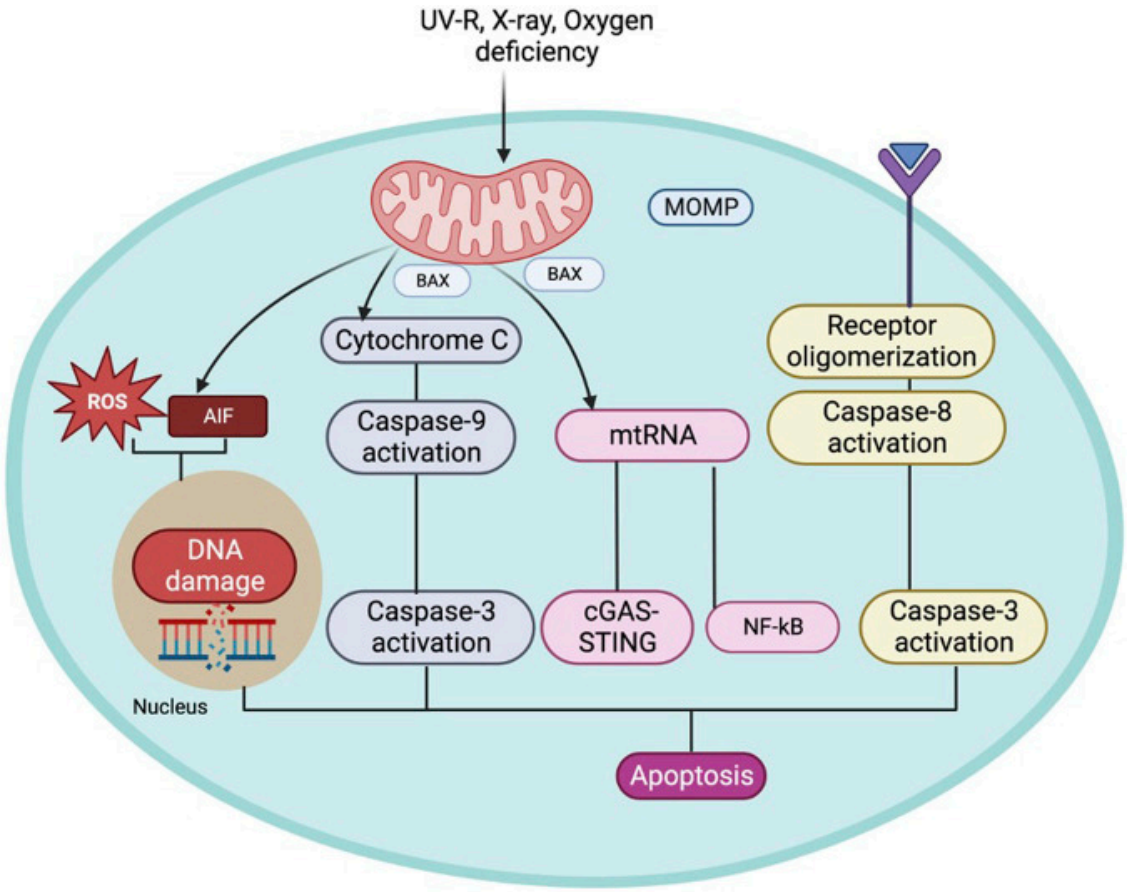
doi:10.3390/biomedicines12081716

Mitochondrial Decline Links Energy Failure, ROS, and Cell Fate



Mitochondrial stress can converge on ROS, mtRNA/mtDNA sensing, NF-kB and apoptosis.

➢ Foundation

Mitochondria generate ATP through OXPHOS and also regulate ROS, calcium signaling, apoptosis and innate immune signaling.

➢ Advanced link

Age-related mitochondrial dysfunction can reduce ATP, increase ROS, release mitochondrial nucleic acids and activate inflammatory pathways such as cGAS-STING/NF-kB.

➢ Care read-out

Energy failure appears as fatigue, low exercise tolerance, slow recovery, delirium vulnerability and lower reserve after acute illness.

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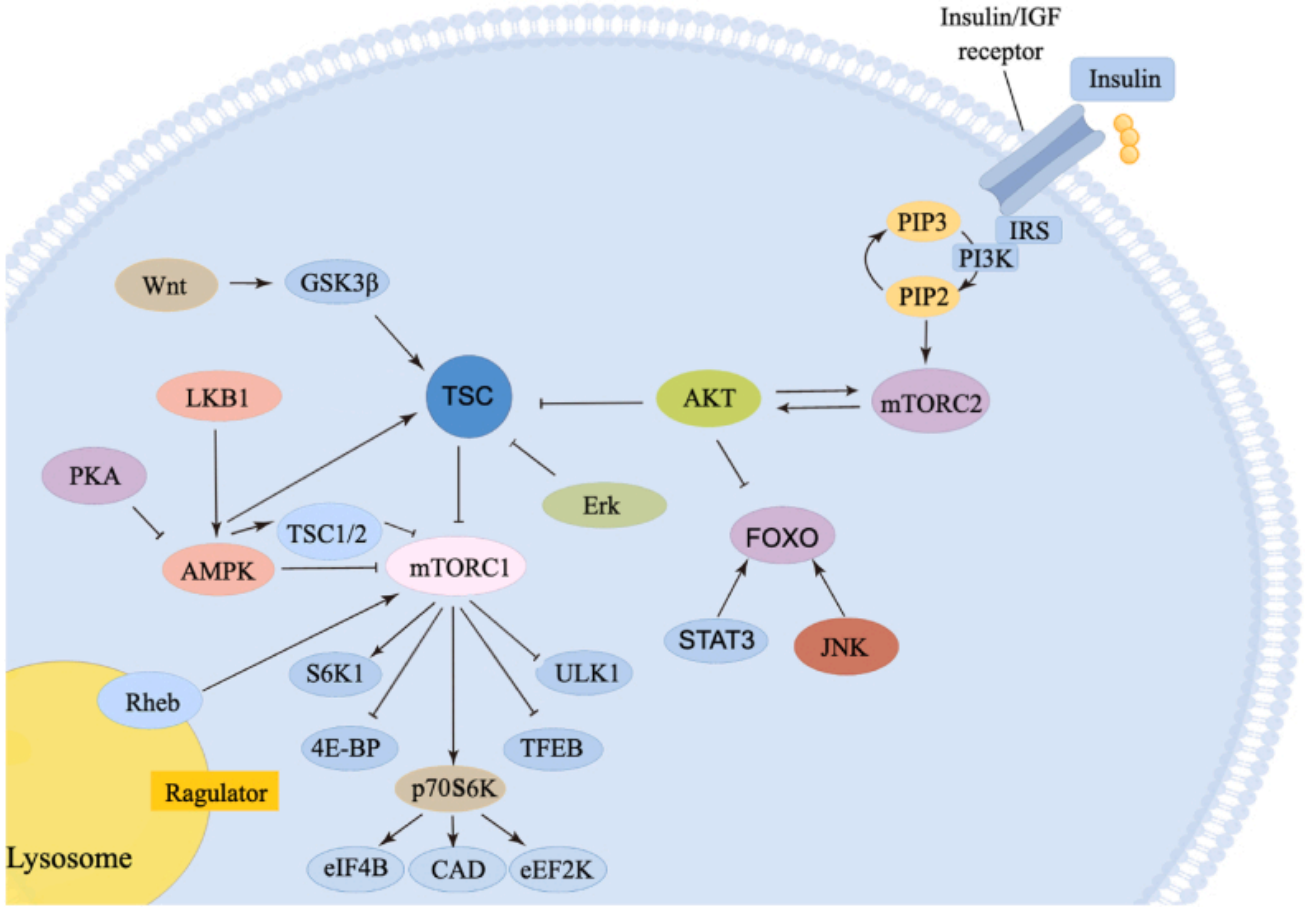
doi:10.3389/fphys.2024.1384966

Nutrient-Sensing Pathways Decide Build, Repair, or Conserve



Insulin/IGF-AKT, AMPK and mTORC1 converge on synthesis, autophagy and metabolism.

Fed / growth state

insulin-IGF (insulin resistance) -> PI3K-AKT -> mTORC1 (upregulation)

protein synthesis, lipid synthesis, growth

Low energy / stress state

AMPK -> TSC/ULK1/TFEB

reduced anabolism, more repair and autophagy

- In ageing, the problem is not one pathway being simply on or off; it is loss of flexible switching.
- **Persistent mTORC1 activity can suppress autophagy and favor accumulation of damaged proteins and organelles.**
- Interventions such as exercise, nutrition timing and some drugs are interpreted through this signaling balance.

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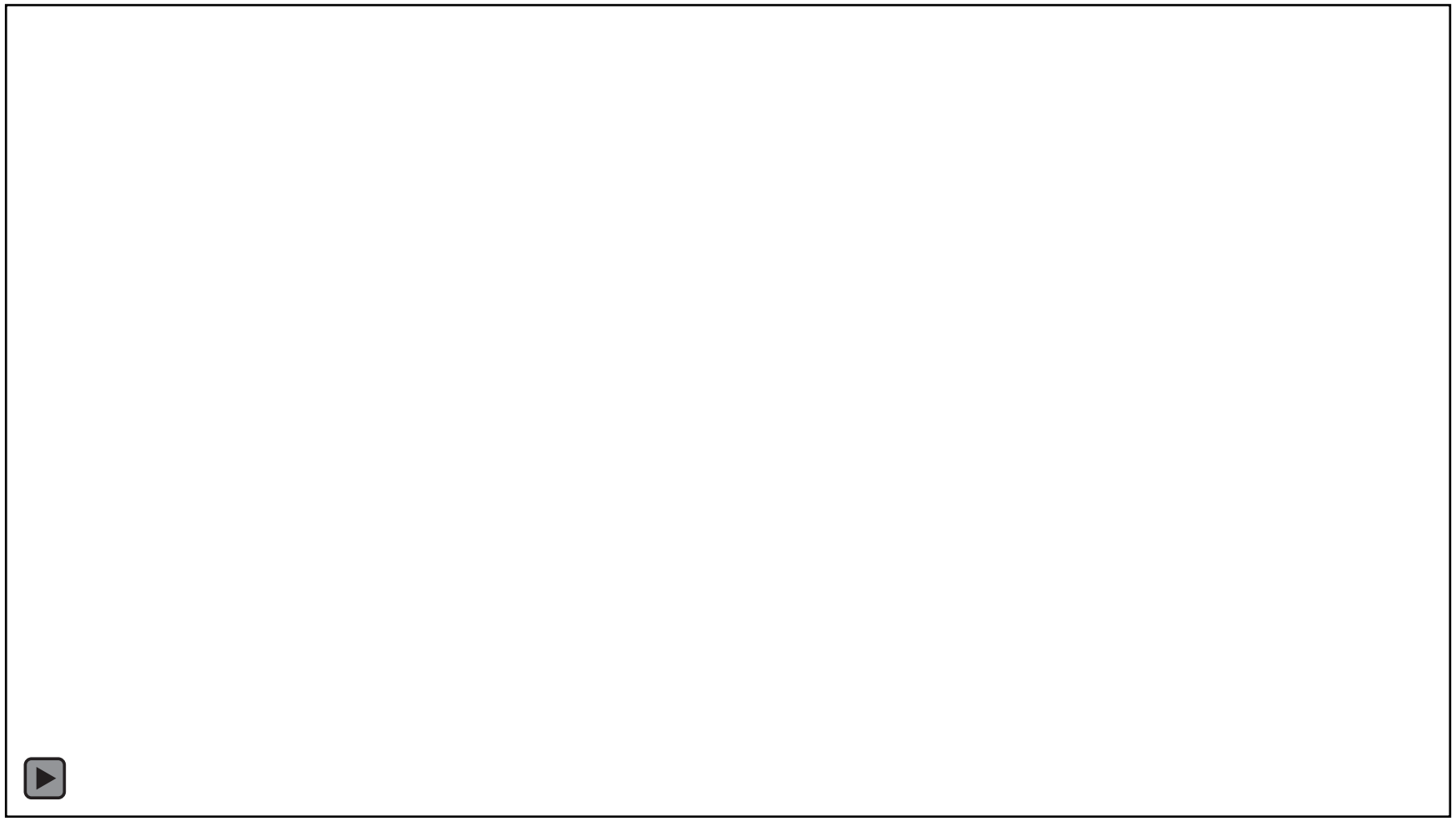
Macao Polytechnic University

doi:10.1016/j.heliyon.2024.e24751

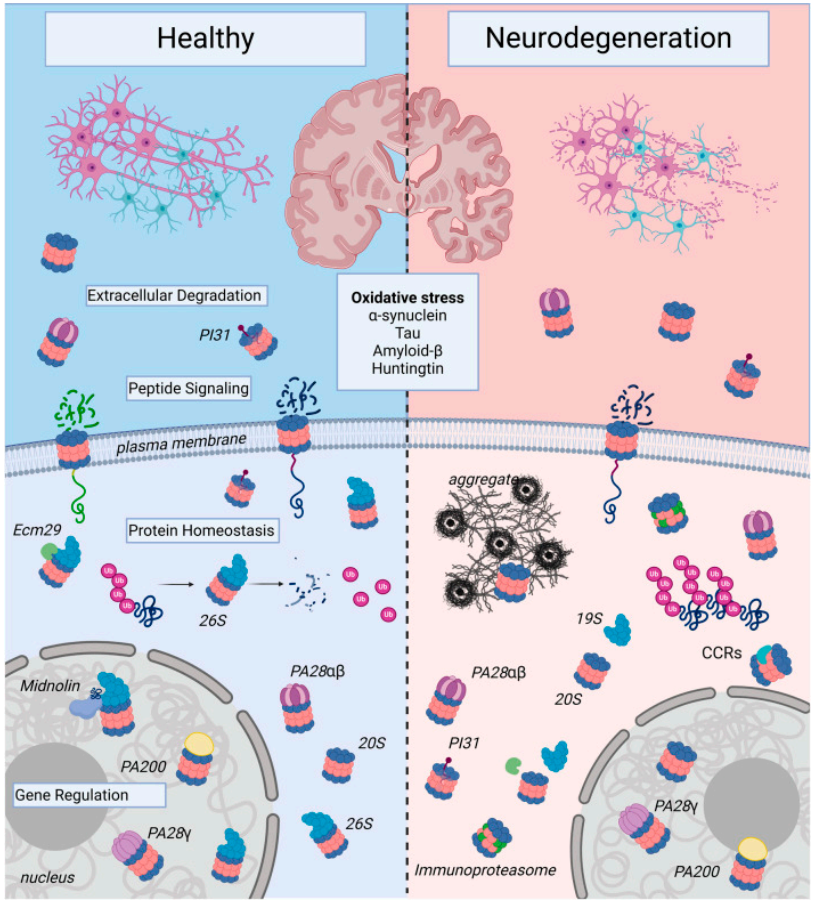
Nutrient-Sensing Pathways Decide Build, Repair, or Conserve

Insulin/IGF-AKT, AMPK and mTORC1 converge on synthesis, autophagy and metabolism.



doi:10.1016/j.heliyon.2024.e24751

Proteostasis Failure Makes Misfolded Proteins Clinically Visible



Proteasome pathways help clear damaged and aggregation-prone neuronal proteins.

Protein homeostasis is a balance between making, folding, refolding, trafficking and degrading proteins.

- **Folding capacity falls**

  chaperones and ER stress responses become less able to rescue misfolded proteins

- **Clearance slows**

  ubiquitin-proteasome and autophagy-lysosome systems lose efficiency

- **Aggregates accumulate**

  tau, alpha-synuclein, amyloid-beta and huntingtin disrupt neurons and synapses

- **Clinical signal appears**

  memory, movement, sleep and behavior become affected before simple cell counts explain the problem

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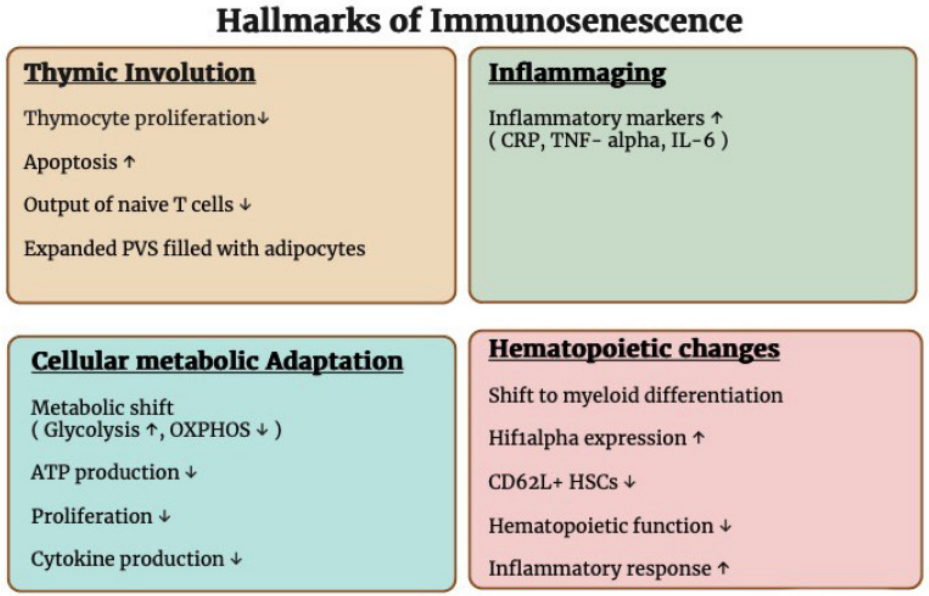
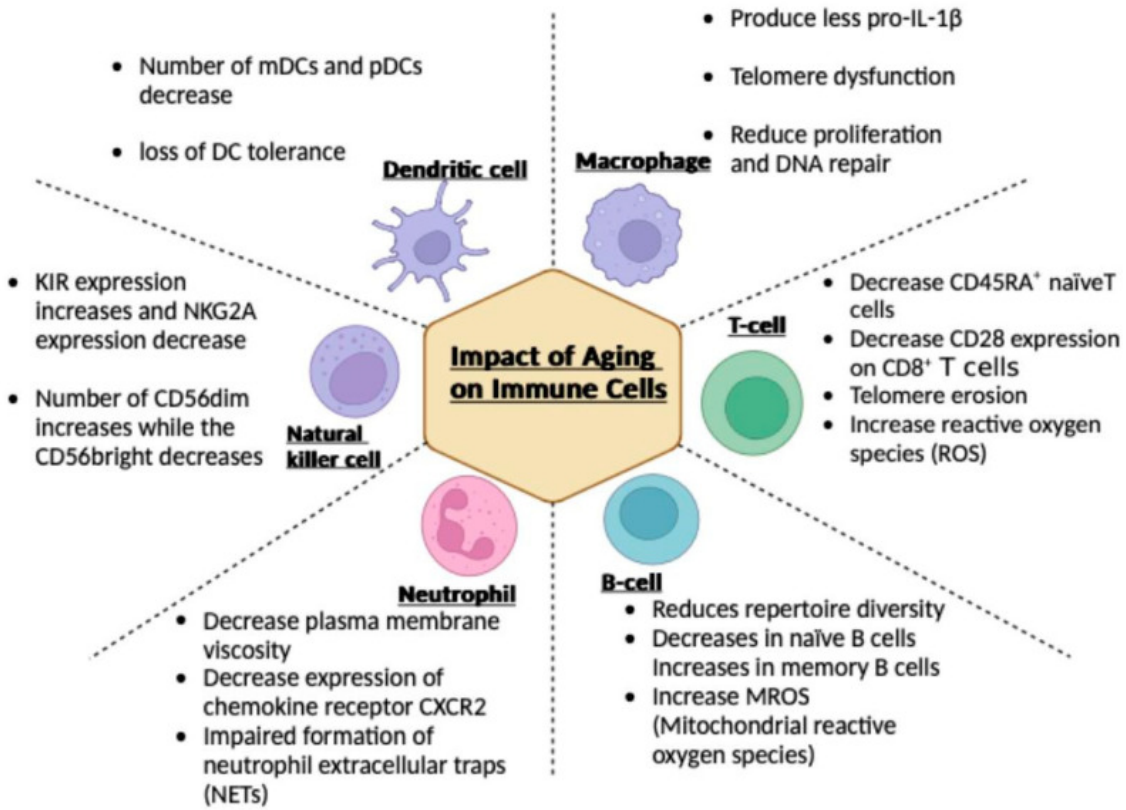
doi:10.3389/fcell.2024.1531797

Immune Ageing Weakens Defense While Raising Basal Inflammation



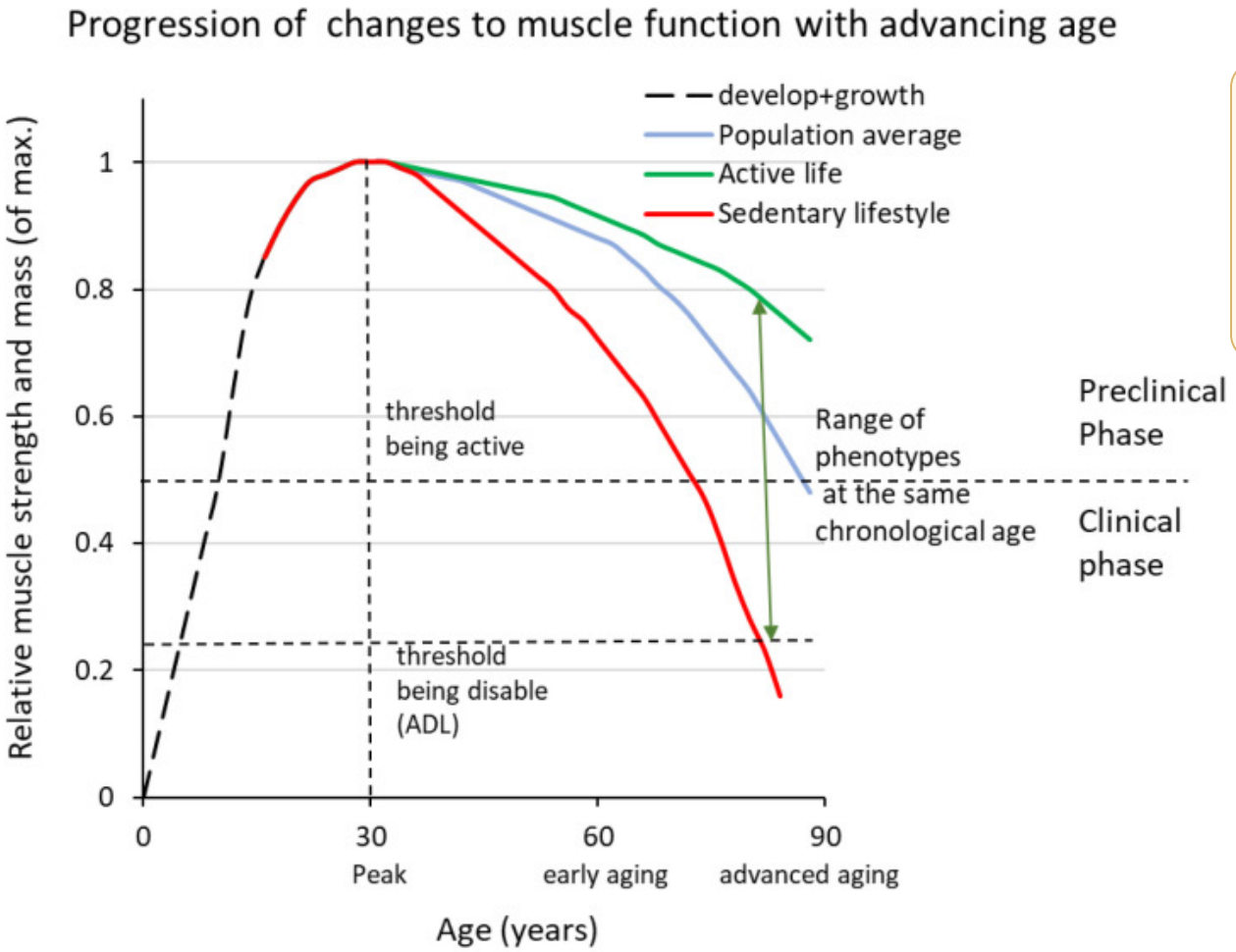
Immunosenescence is not simply fewer immune cells; it is a remodeled immune system with less naive diversity, altered cell metabolism and more inflammatory tone.



- **T-cell output declines with thymic involution**, while memory and late-differentiated cells occupy more immune space.
- **B-cell repertoire diversity and germinal-center responses** can weaken, reducing antibody quality after vaccination.
- **Innate cells may show impaired** pathogen clearance yet maintain pro-inflammatory signaling.
- Care implication: infection prevention, vaccine timing, nutrition, sleep and medication review become part of immune support.

doi:10.3390/vaccines12121314

Sarcopenia Begins Before Disability is Obvious



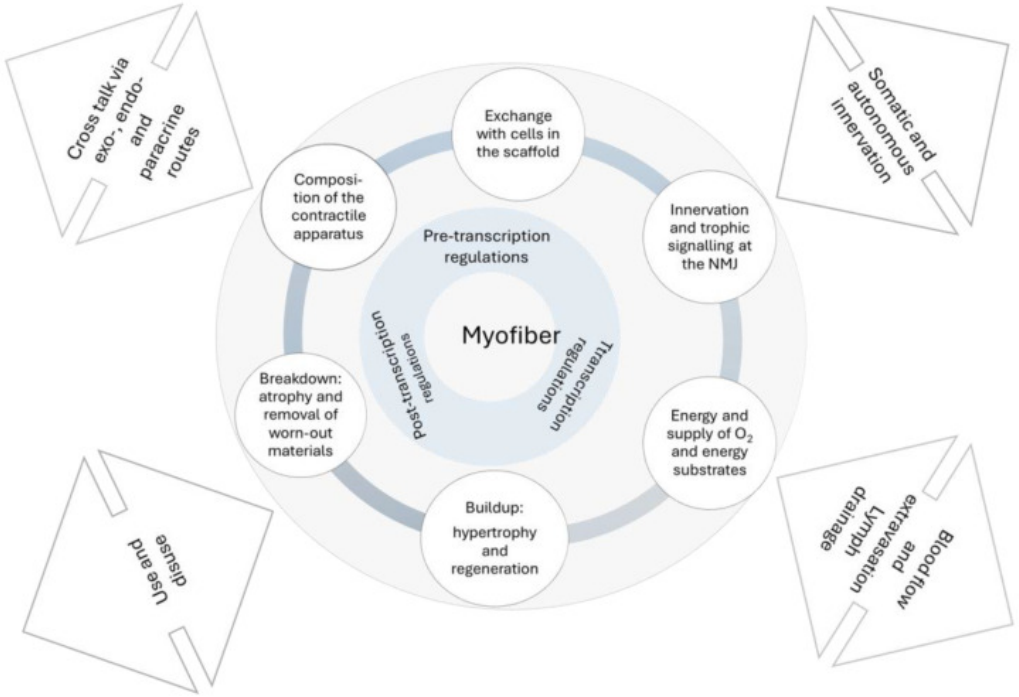
Activity can shift the trajectory before clinical sarcopenia becomes obvious.

Muscle ageing is a multi-compartment problem: myofibers, motor units, neuromuscular junctions, mitochondria, satellite cells and inflammatory signals all matter.

- **Strength and power often decline before** muscle mass alone crosses a diagnostic threshold.
- Motor-unit loss and incomplete reinnervation reduce fine control and increase fall risk.
- Reduced protein synthesis response, mitochondrial capacity and repair capacity limit recovery after illness or bed rest.
- **Care needs: resistance exercise, protein adequacy, fall prevention, medication review and recovery planning.**

doi:10.3390/ijms252010932

Mechanisms Become Useful Through an Organ-Function-Care Chain



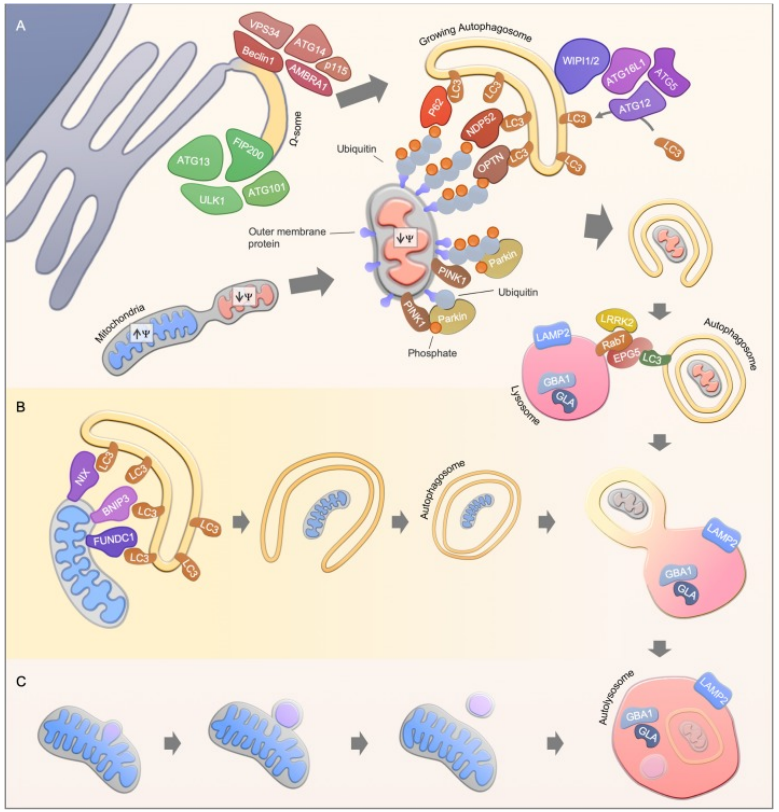
Organ vulnerability depends on cells, scaffold, innervation and energetic supply.

Doctoral-level integration means asking not only 'which pathway changed?' but also 'which measurable function and care decision should this explain?'

Mechanism	>	Organ	>	Function	>	Care need
senescence mitochondria mTOR/proteostasis immune ageing		muscle brain lung vascular system immune organs		mobility memory sleep infection defense ADL capacity		prevention monitoring rehab vaccination home support

adapted from doi:10.3390/ijms252010932; 10.1016/j.cell.2022.11.001

Advanced Mechanisms Should Change How We Assess and Support Ageing



Mechanistic depth is useful when linked to phenotype, timing and intervention.

➢ **Mechanistic marker**

SASP cytokines, immune subsets, mitochondrial signals, proteostasis stress

➢ **Functional phenotype**

gait speed, grip strength, ADL/IADL, cognition, sleep, fatigue, vaccine response

➢ **Context signal**

nutrition, activity, medication burden, infection exposure, living environment

➢ **Care action**

risk stratification, prevention, rehabilitation intensity, monitoring, shared care planning

The central move: translate pathway biology into care-relevant models of reserve, risk, recovery and support.

doi:10.3389/fcell.2020.00239; 10.1016/j.cell.2022.11.001

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Universidade Politécnica de Macau
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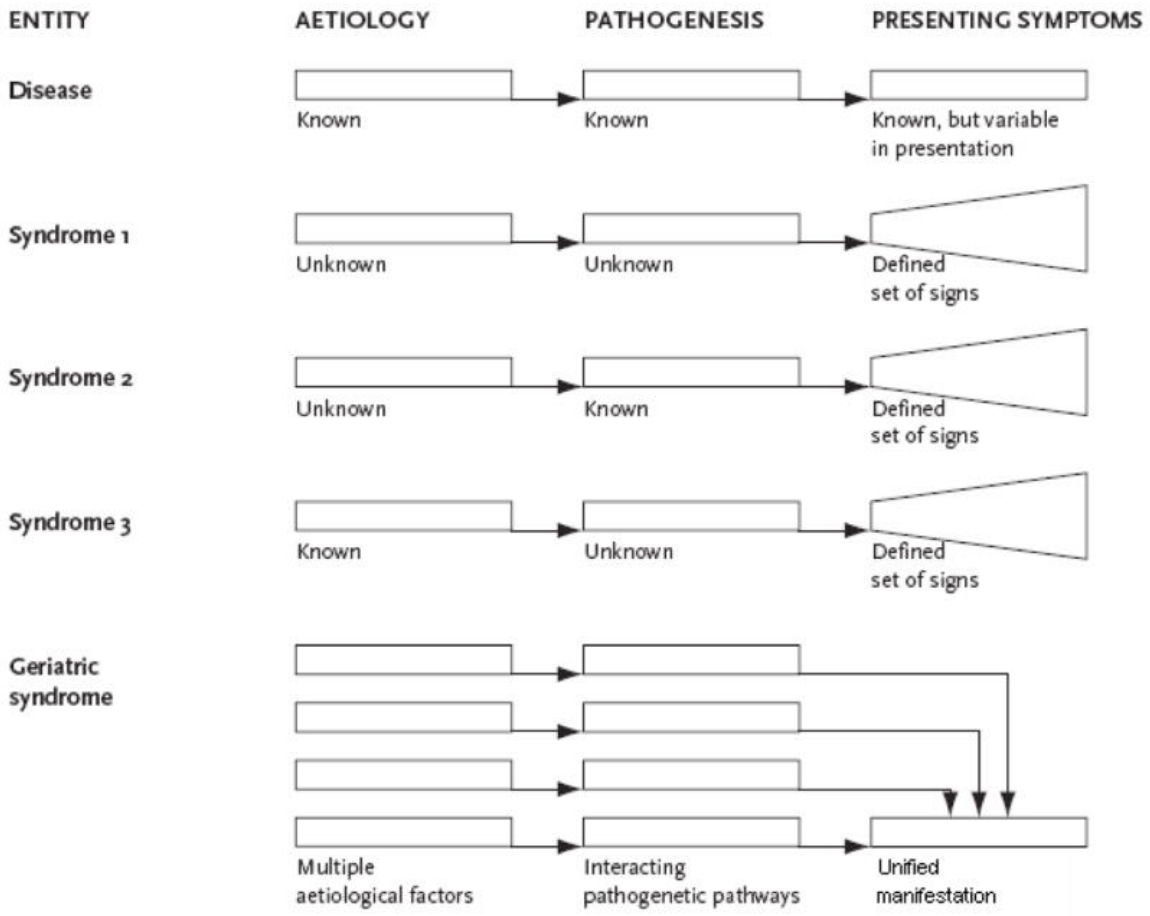
Geriatric Syndromes: Foundations

Seeing patterns that single-disease thinking misses

A geriatric syndrome is a functional warning sign: multiple vulnerabilities meet a stressor, and the result is often falls, delirium, incontinence, pressure injury or decline.

doi:10.1111/j.1532-5415.2007.01156.x

A Geriatric Syndrome is a Multifactorial Functional Phenotype



The geriatric syndrome column expands causes, mechanisms and manifestations.

Working definition

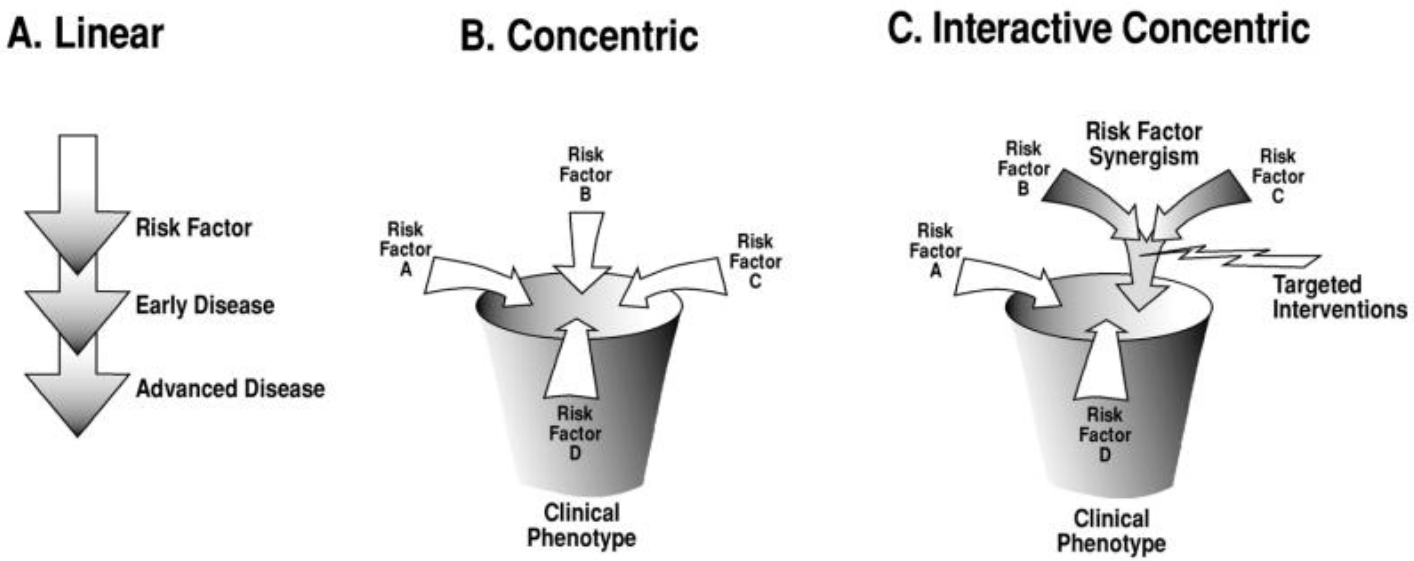
A multifactorial condition in which accumulated impairments across systems make an older person vulnerable to situational challenges.

- The syndrome is observed as a clinical event or functional change, not as one isolated lesion.
- The same visible outcome can arise from different combinations of risks.
- The same risk factor can contribute to several syndromes at once.
- Assessment must ask what changed in function, reserve and context.

Do not start with a single organ. Start with a vulnerable person in a real situation.

doi:10.1111/j.1532-5415.2007.01156.x

Single-Disease Logic Breaks Down When Risks Interact



Interactive models are needed when causes reinforce each other rather than form one line.

Single-disease model

One main pathology -> one organ -> one treatment target

Geriatric-syndrome model

Many small deficits + stressor + environment -> functional event

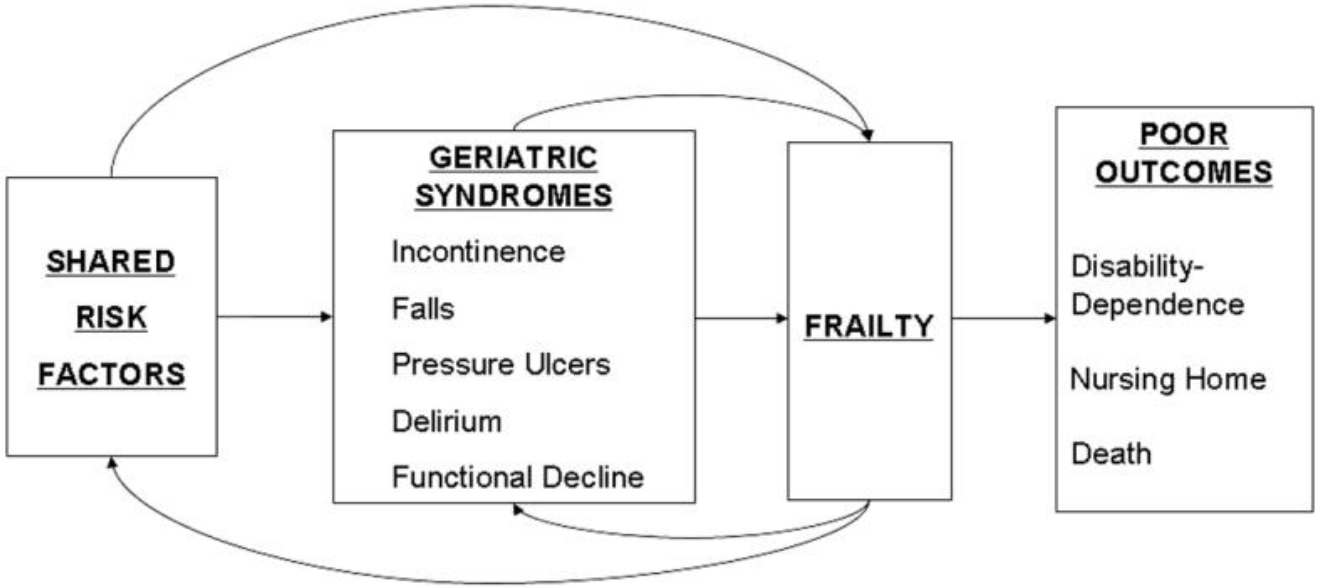
Clinical shift

Move from naming disease only to estimating reserve, thresholds, triggers and recovery.

- **Falls may involve cognition, vision, medications,** hypotension, strength, footwear and home hazards.
- Delirium may be triggered by infection, pain, sleep loss, dehydration or medication in a vulnerable brain.
- **Functional decline often marks the combined effect of disease, immobility, nutrition and care environment.**

doi:10.1111/j.1532-5415.2007.01156.x

Shared Risk Factors Explain Why Syndromes Cluster



Cognitive impairment, mobility impairment, disability and age act across multiple syndromes.

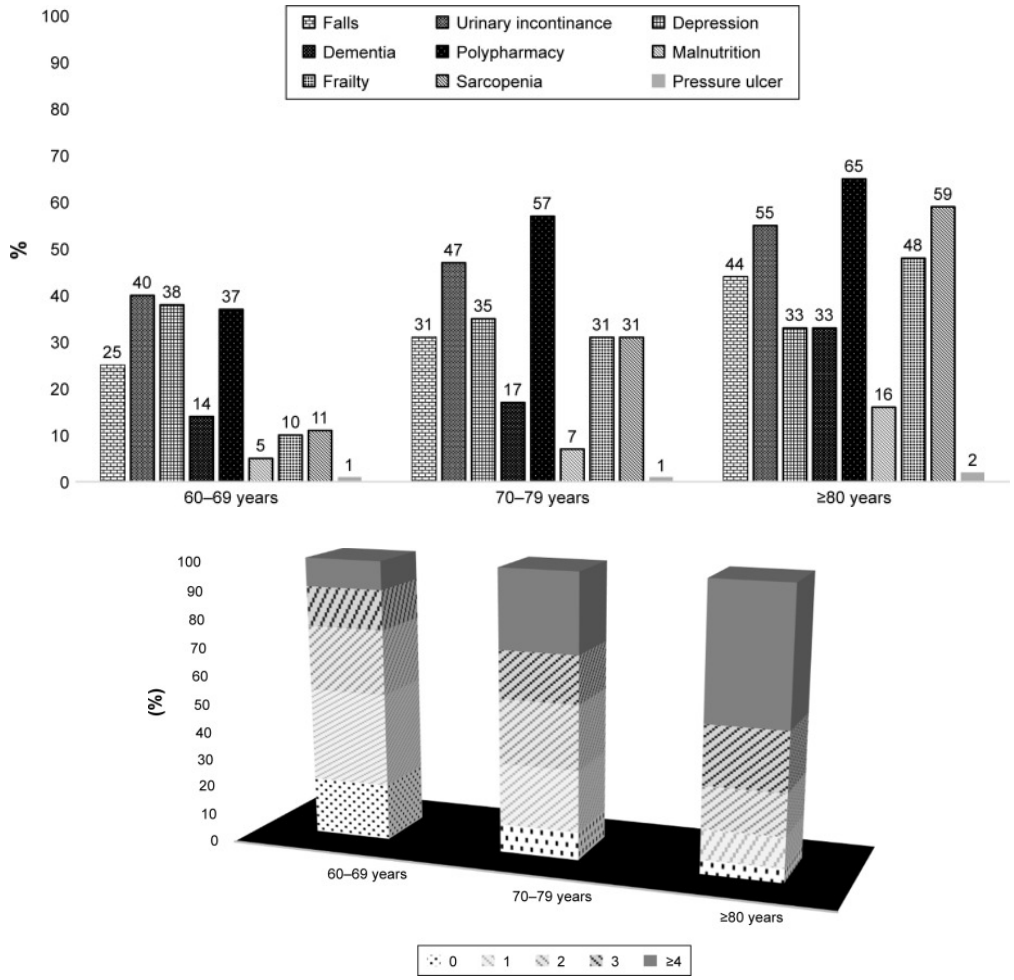
A shared risk factor is useful because changing it may reduce several outcomes at once.

- Risk factors are not merely predictors; they are often intervention targets.
- The same older person may move between syndromes as stressors change.
- Frailty can be viewed as both a syndrome and a system-level vulnerability.

Cognition	Mobility	Function	Age	Multimorbidity
attention, memory, executive control	gait, balance, strength	ADL/IADL, reserve	physiology, exposure time	disease burden, polypharmacy

doi:10.1111/j.1532-5415.2007.01156.x

Frailty, Falls, Delirium and Other Syndromes Often Co-Occur



Population data show that syndromes become more frequent and more cumulative with age.

Frailty
low reserve,
slow recovery

Falls
mobility + context
+ medication

Delirium
acute brain failure
under stress

Incontinence
bladder, mobility,
cognition, drugs

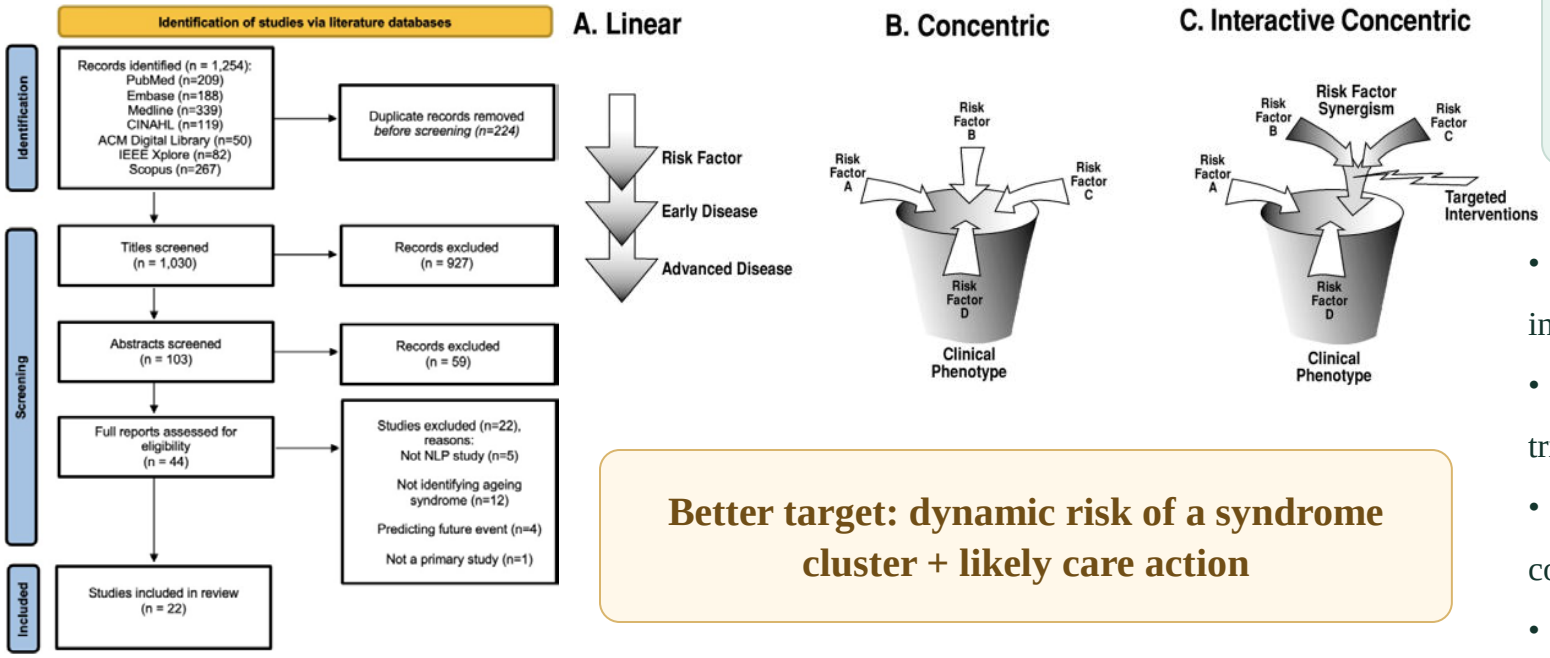
Pressure injury
immobility,
nutrition, perfusion

Functional decline
loss of ADL/IADL
capacity

The clinical question is not 'which one label is correct?' but 'which cluster is driving risk today?'

doi:10.2147/CIA.S180281

Geriatric Syndromes Are Hard Targets for AI Models



Better target: dynamic risk of a syndrome cluster + likely care action

Evidence synthesis and conceptual models both point to sparse labels and interacting causes.

The modeling problem is not only prediction; it is representing vulnerability, triggers and recovery over time.

- **Labels are inconsistent**: falls, delirium, frailty and incontinence **may be absent, delayed or hidden in notes**.
- **Events are time-dependent**: predisposing risks and acute triggers must be separated.
- **Features interact**: cognition, mobility, medications, comorbidity and environment are not independent inputs.
- **Validation is fragile**: a 2024 NLP review found only 4 of 22 studies used external validation.
- **Care usefulness requires actionability, not just an AUC.**

doi:10.1093/ageing/afae135; 10.1111/j.1532-5415.2007.01156.x

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Priority Geriatric Syndromes

From recognizable syndromes to care decisions

Syndromes are where mechanisms, chronic disease, environment and care systems become visible as daily-life risk.

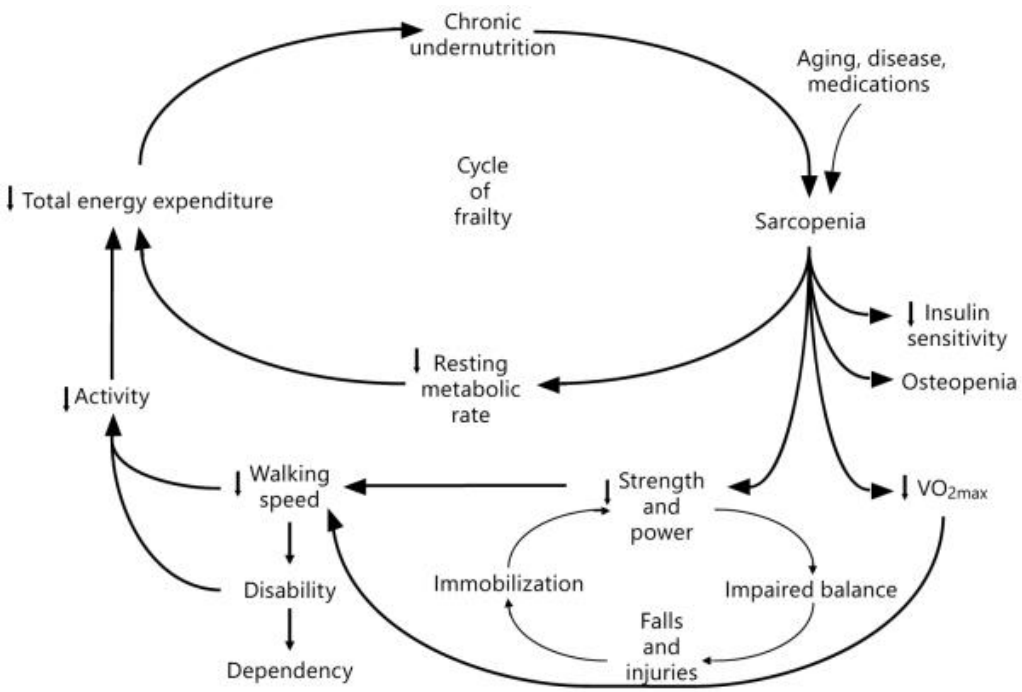
684,000
annual deaths from falls

57M
living with dementia in 2021

1 in 6
people worldwide report loneliness

WHO: Step Safely; dementia fact sheet; social connection commission

Frailty Has Two Useful Models, Not One



Frailty cycle: weight loss, weakness, exhaustion, inactivity and slowed performance reinforce one another.

➢ **Phenotype model**

A physical syndrome defined by observable criteria: shrinking, weakness, exhaustion, slowness and low activity.

➢ **Deficit accumulation model**

A frailty index sums symptoms, diseases, disabilities and abnormalities to estimate vulnerability.

- Both models ask about reserve: how much stress can this person tolerate before function collapses?
- Phenotype is intuitive for bedside screening; deficit accumulation is flexible for EHR and cohort data.
- Frailty is not simply old age. It is a measurable state of low resilience and slow recovery.

Low reserve > **Minor stressor** > **Functional loss** > **Care need**

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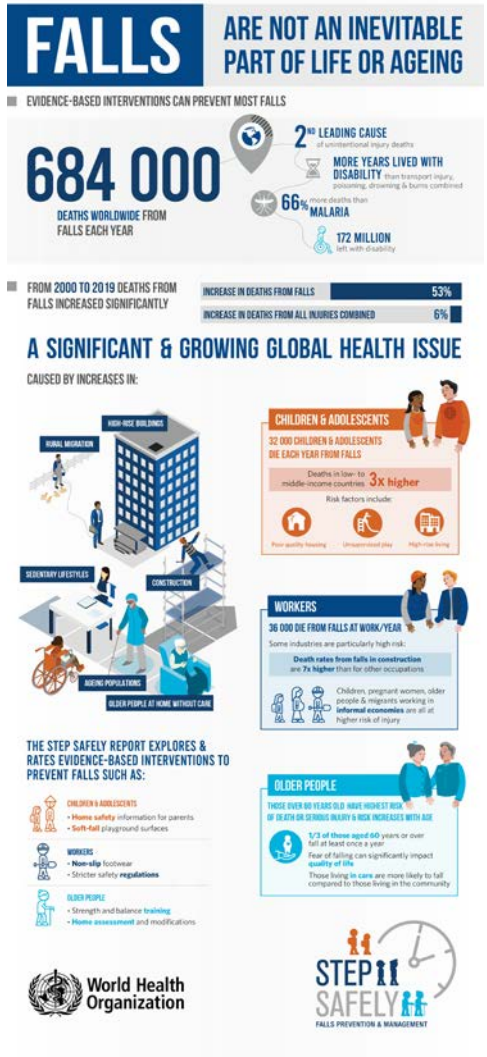
doi:10.1159/000381164; 10.1093/gerona/56.3.m146

Falls Happen When Person, Task and Environment Misalign



684,000
people die from falls each year

Intrinsic gait and balance weakness

vision or hearing loss

orthostatic hypotension

cognition or fear of falling

172M
people experience fall-related disability

Extrinsic sedatives and antihypertensives

footwear and assistive devices alcohol or acute illness rushed transfers

Preventable exercise, medication review and safer environments

Environment

stairs and bathrooms

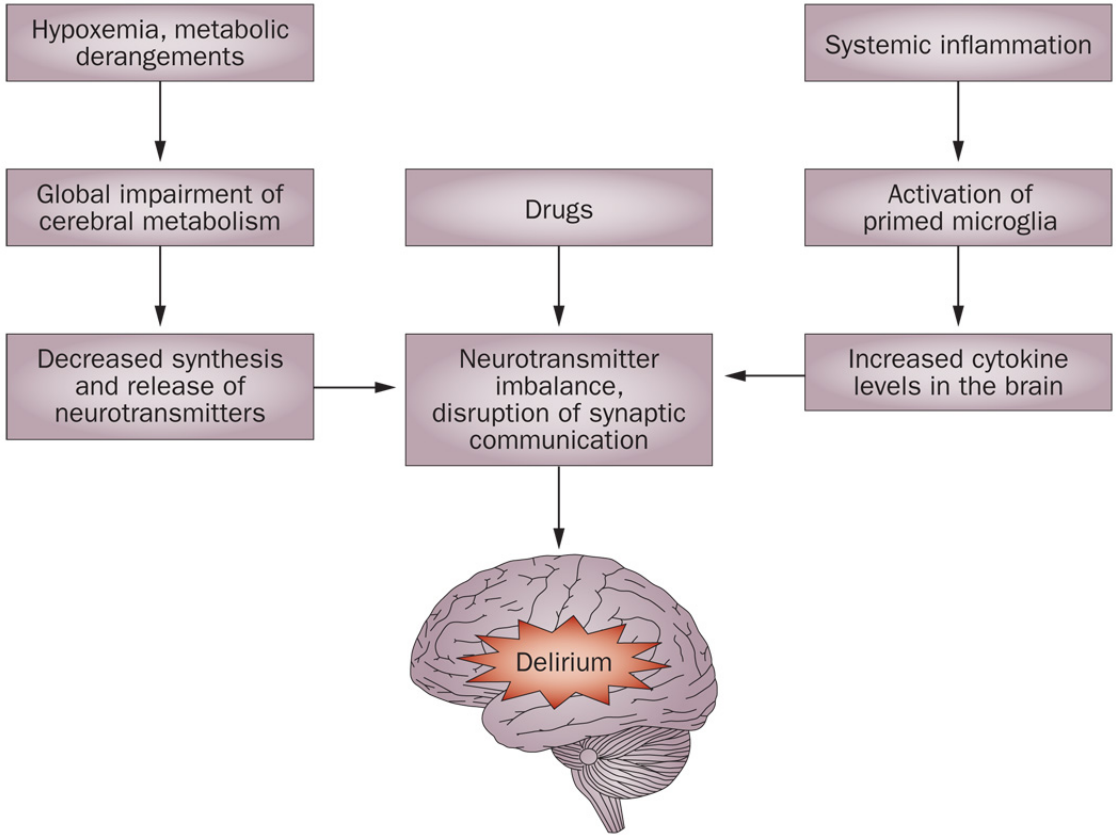
poor lighting or clutter uneven ground lack of handrails

Care translation: screen gait and balance, review medicines, test vision and orthostasis, modify the home, and prescribe strength + balance training.

WHO Step Safely frames falls as preventable across settings and life-course stages.

WHO Step Safely; guideline doi:10.1093/ageing/afac205

Delirium is Acute Brain Failure, Usually With a Trigger



Predisposing vulnerability plus acute insults disturb neurotransmitters, inflammation and brain networks.

Beginner definition

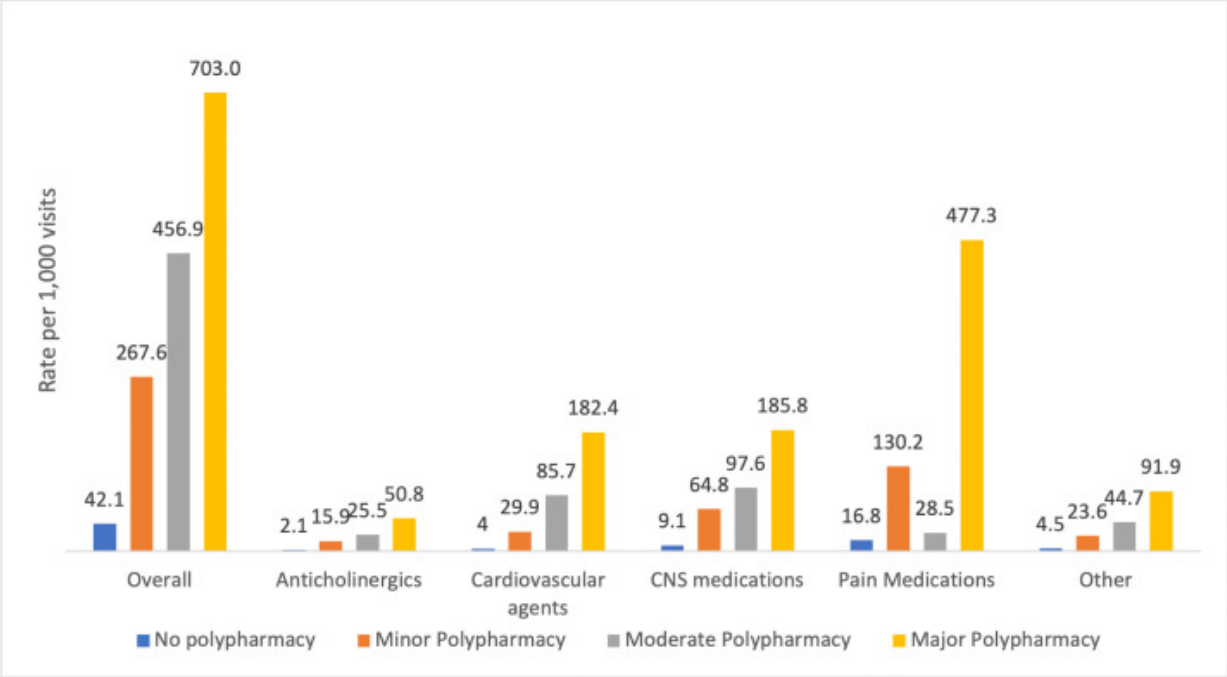
Delirium is an acute, fluctuating disturbance of attention, awareness and cognition.

- It often appears after infection, surgery, dehydration, hypoxia, pain, sleep loss or medication changes.
- It is potentially reversible, but delayed recognition can lead to persistent cognitive and functional decline.
- Hypoactive delirium is easy to miss because the person may look quiet rather than agitated.
- Prevention bundles target orientation, sleep, mobility, hydration, sensory aids and medication burden.

Clinical habit: when cognition changes over hours to days, search for a precipitant before calling it dementia.

doi:10.1038/nrneurol.2009.24

Medication Burden Turns Treatment into a Risk Factor



High-risk medication categories illustrate why number of drugs is only one part of burden.

Polypharmacy is not just a count.

The burden comes from high-risk medicines, interactions, cumulative anticholinergic or sedative effects, adherence workload and prescribing cascades.

- **CNS-active drugs** increase falls, delirium and cognitive slowing.
- Anticholinergic load can worsen constipation, urinary retention, confusion and dry mouth.
- Guidelines for one disease may conflict with goals for frailty, dementia or limited capacity.
- Deprescribing starts with goals, benefit horizon, withdrawal risk and patient preference.

More drugs > **More burden** > **Lower function** > **More syndromes**

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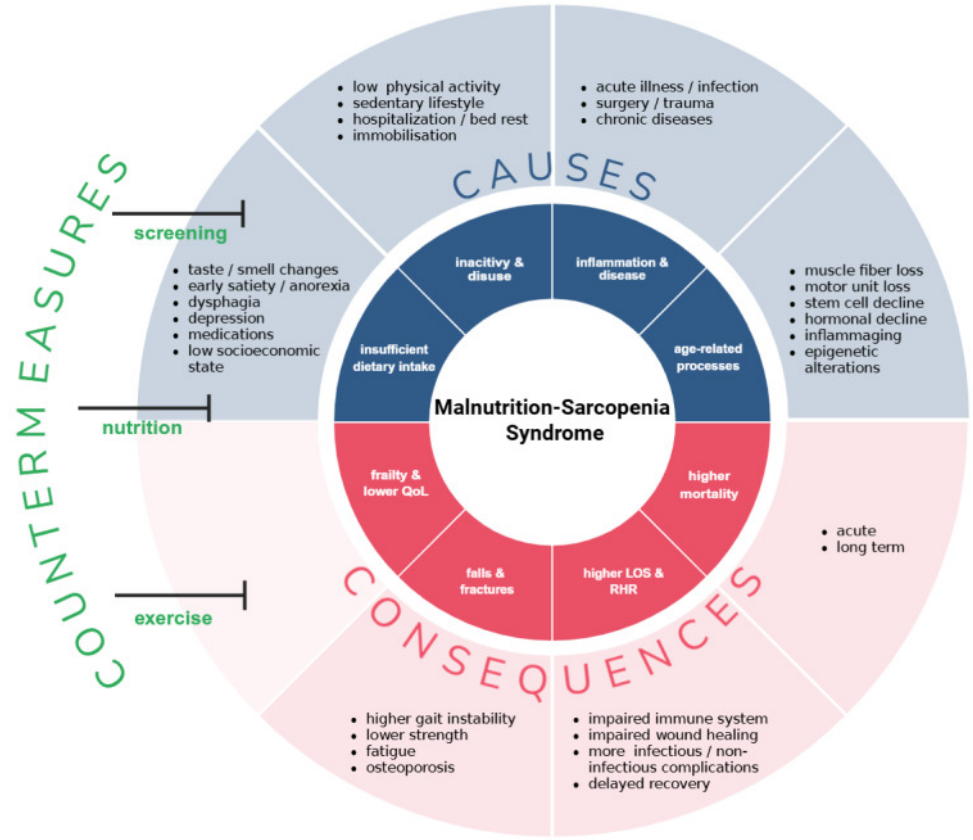
doi:10.1371/journal.pone.0255642; 10.1111/jgs.15767

Sarcopenia and Malnutrition Reinforce Each Other



Malnutrition-sarcopenia syndrome links causes, consequences and countermeasures in older adults.

➢ Sarcopenia

Loss of muscle strength, quantity or quality, and physical performance.

➢ Malnutrition

Insufficient energy, protein or micronutrients that reduce tissue repair and resilience.

- **Low intake, inflammation, illness and inactivity reduce muscle protein synthesis.**
- **Weakness lowers mobility and appetite, increasing falls, dependence and hospitalization risk.**
- Screen with weight loss, appetite, grip strength, gait speed, calf circumference and nutrition tools.
- Core intervention: adequate protein/energy plus progressive resistance exercise; treat oral, swallowing and disease drivers.

Care target: rebuild capacity, not only increase calories.

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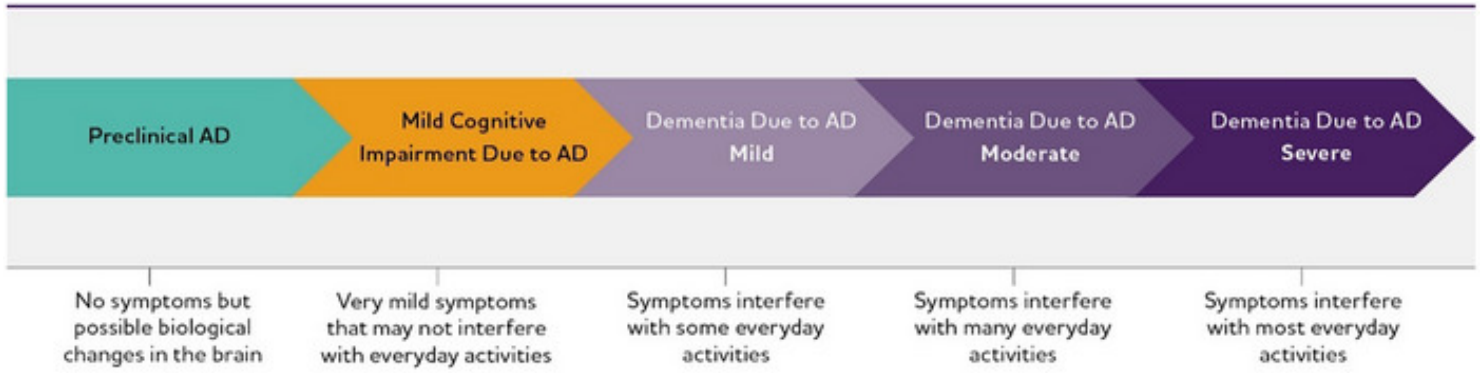
doi:10.17179/excli2026-9426; 10.1093/ageing/afy169

Cognitive Decline Becomes a Care Issue Before Dementia is Named



Cognitive decline may move from silent pathology to MCI and then dementia with loss of independence.

57M
people living with dementia worldwide in 2021

Nearly 10M
new cases every year

>60%
live in low- and middle-income countries

- Distinguish acute change from delirium and chronic progressive change from dementia.
- Look for reversible contributors: hearing or vision loss, sleep disorders, depression, drugs, thyroid/B12 problems and infections.
- Measure function as carefully as cognition: ADL/IADL change often determines care need.
- Support the caregiver system, because dementia risk is also household risk.

➢ **Why it matters for care**

Memory loss becomes a syndrome when it changes medication safety, finances, cooking, navigation, sleep, behavior, caregiver workload and risk of exploitation.

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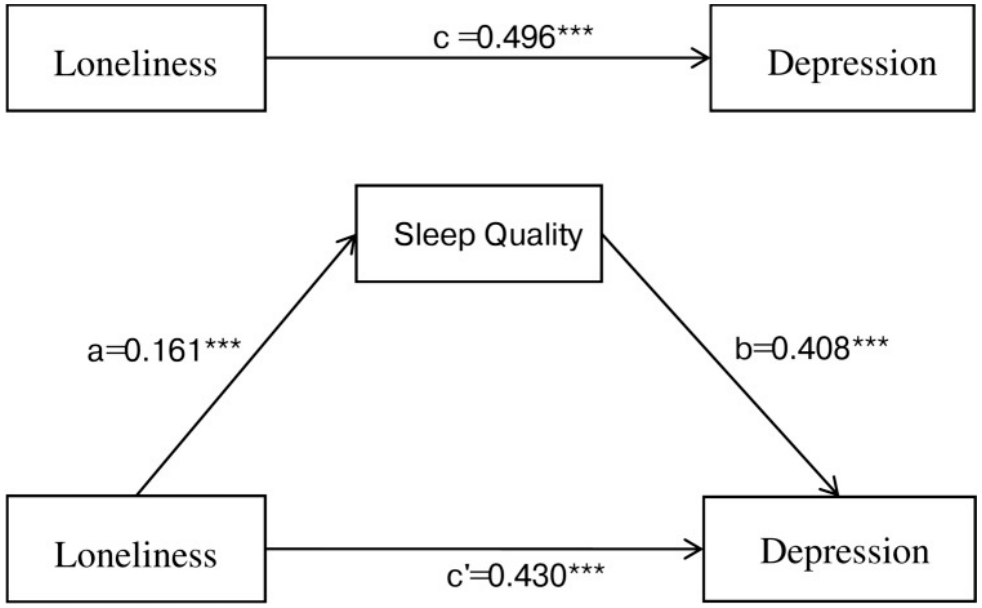
doi:10.1002/alz.13809; WHO dementia fact sheet

Depression, Loneliness and Poor Sleep Form a Feedback Loop



A mediation analysis links loneliness, poor sleep and depressive symptoms in older adults.

➢ **Beginner frame**

Mood, sleep and social connection are not soft outcomes. They change appetite, activity, pain perception, cognition, inflammation and treatment adherence.

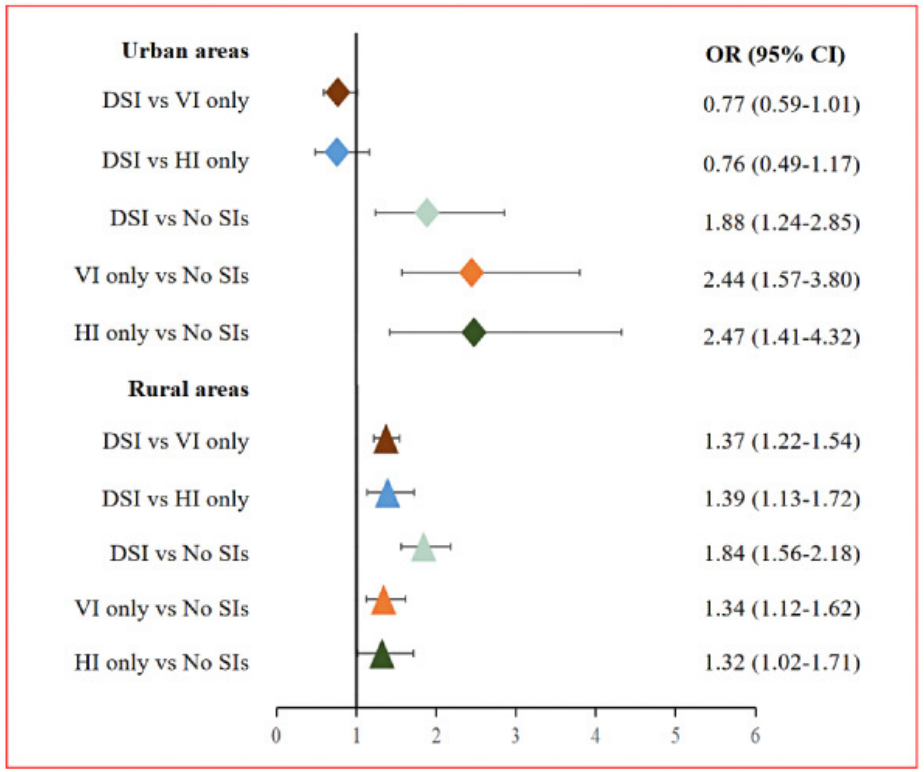
1 in 6 people worldwide experience loneliness

11.8% older people report loneliness

- Loneliness may be a cause, marker and consequence of illness.
- Poor sleep can amplify mood symptoms, falls risk, pain and cognitive complaints.
- Assessment should ask about bereavement, living alone, hearing loss, mobility limits and digital exclusion.
- Interventions combine social prescribing, sleep hygiene, activity, pain control and depression treatment.

doi:10.1038/s41598-025-93681-3; WHO social connection

Pain and Sensory Loss Push People Out of Participation



Hearing, vision and dual sensory impairment are associated with loneliness in older adults.

- **The pathway is often indirect**

Pain or sensory loss reduces movement, communication confidence and perceived safety. The result may look like withdrawal, depression or cognitive decline.

- Chronic pain narrows activity and sleep, and can increase fear of falling.
- Hearing loss makes conversation effortful; vision loss changes mobility and medication safety.
- Withdrawal may be a rational adaptation to an unsafe or exhausting environment.
- Care requires pain assessment, hearing/vision correction, environmental adaptation and communication support.

Pain or sensory loss > **Less activity** > **Social withdrawal** > **Lower function**

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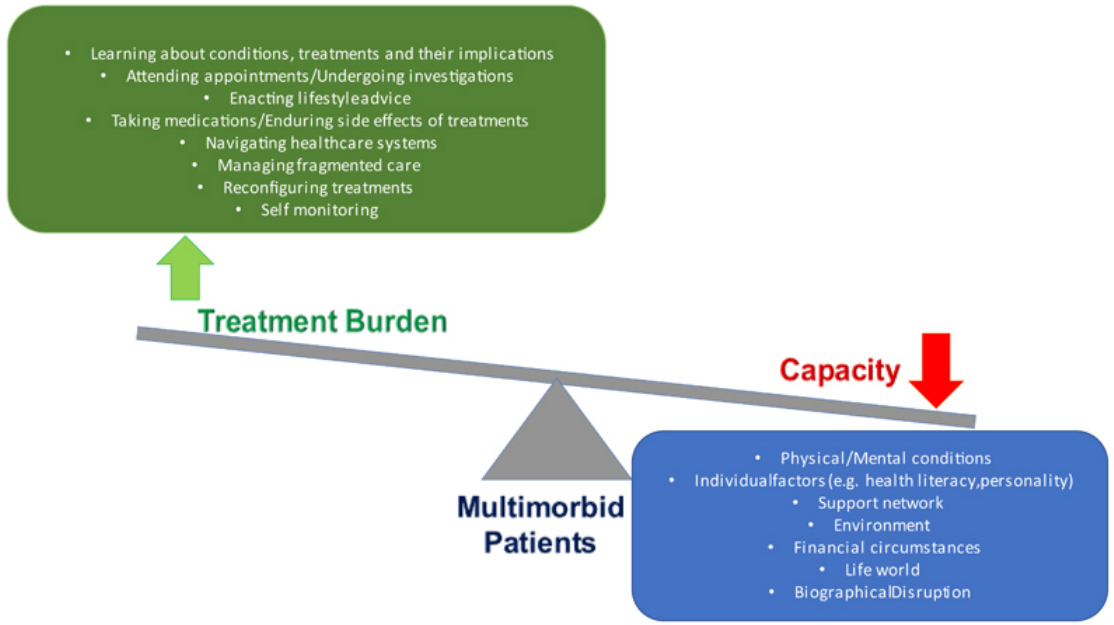
doi:10.2196/39314

Multimorbidity Amplifies Syndromes Through Burden and Low Capacity



When treatment workload exceeds capacity, care itself can become part of the problem.

➢ **Multimorbidity rises with age**

Approximate prevalence in reviewed evidence: 30% at 45-64 years, 65% at 65-84 years, and 82% at 85+ years.

- Disease-disease interaction: heart failure, arthritis, diabetes and dementia can each worsen functional reserve.
- Disease-treatment interaction: treatment for one condition may worsen falls, delirium, continence or nutrition.
- Treatment-treatment interaction: drugs, appointments and monitoring compete for time, money and attention.
- Social context determines capacity: caregiver support, transport, housing, literacy and income change what is feasible.

Teaching synthesis: prioritize what most threatens function today, then reduce burden while rebuilding capacity.

doi.10.1038/s41572-022-00376-4

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Functional Status Assessment

Turning daily ability into care priorities

CGA asks a practical question: what can this older person do today, what is becoming unsafe, and what support would preserve independence?

Assess ability > **Find risks** > **Set goals** > **Plan support**

doi:10.1007/s41999-026-01420-2; WHO ICOPE

ADL and IADL Separate Self-Care From Independent Living

ADL: basic self-care
Can the person keep the body safe, clean, fed and mobile within the home?

ADL Domain	What To Ask
Eating	Needs help cutting food, feeding or swallowing?
Dressing	Can choose clothes and put them on safely?
Bathing	Can wash without falls or exhaustion?
Toileting	Can reach, transfer, clean and manage continence?
Transferring	Can move bed-chair-standing without help?

IADL: complex daily management
Can the person manage the household, community tasks and decisions needed for independent living?

IADL Domain	What To Ask
Shopping	Can plan, travel, pay and carry items?
Cooking	Can prepare meals and use appliances safely?
Medication	Can understand, organize and take doses?
Finances	Can pay bills and avoid mistakes or fraud?
Transport	Can use car, taxi, bus or assisted transport?

Interpretation: ADL loss often signals personal care need; IADL loss often appears earlier and signals risk for unsafe living alone.

doi:10.1001/jama.1963.03060120024016; 10.1093/geront/9.3_Part_1.179

Mobility, Strength and Activity Reveal Functional Reserve

1

Gait Speed
Usual walking speed over a short distance.
Slow speed is a global marker of low reserve.

2

TUG
Stand up, walk 3 m, turn, return and sit.
Captures transfer, gait, turning and balance.

3

Balance
Side-by-side, semi-tandem or tandem stance.
Shows postural control and fall risk.

4

Grip Strength
Simple proxy for overall muscle strength.
Useful when full performance testing is hard.

5

Physical Activity
How much the person actually moves.
Connects capacity to daily behavior.

6

Beginner interpretation
Mobility tests are not athletic tests. They estimate how close a person is to the threshold where small stressors become falls, hospitalization or dependence.

Capacity > **Performance** > **Daily activity** > **Risk threshold**

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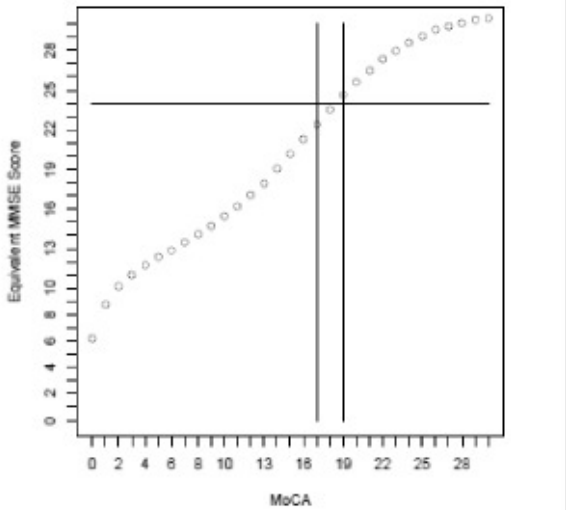
doi:10.1111/j.1532-5415.1991.tb01616.x; 10.1001/jama.2010.1923

Cognition, Mood and Support Explain Why Function Changes

MoCA	Equivalent MMSE	MoCA	Equivalent MMSE
0	6	16	21
1	9	17	22
2	10	18	24
3	11	19	25
4	12	20	26
5	12	21	27
6	13	22	27
7	14	23	28
8	14	24	29
9	15	25	29
10	15	26	30
11	16	27	30
12	17	28	30
13	18	29	30
14	19	30	30
15	20		



MoCA and MMSE are related but not interchangeable; interpretation depends on the clinical question.

- A low score tells you to investigate; it does not explain the cause.
- Function helps separate mild cognitive impairment from dementia.
- Collateral history is essential when insight, memory or safety is uncertain.
- Mood and social isolation are modifiable contributors to functional decline.

➢ **Cognitive screening**

MMSE and MoCA are brief screens, not stand-alone diagnoses. Use them to structure attention, memory, orientation, language, visuospatial and executive-function questions.

➢ **Mood and sleep**

Depression, anxiety, poor sleep and grief may reduce motivation, concentration and activity, mimicking or worsening cognitive decline.

➢ **Social support**

Ask who notices change, who helps with decisions, and whether the person can call for help when routines fail.

doi:10.1186/s12877-015-0103-3; 10.1111/j.1532-5415.2005.53221.x

Home and Caregiver Assessment Turn Scores Into a Care Plan

➢ **Home environment**
Lighting, bathroom safety, stairs, clutter, kitchen hazards, medication storage, emergency access and distance to services.

➢ **Caregiver burden**
Time, emotional strain, sleep disruption, financial pressure, confidence, health of caregiver and backup support.

➢ **Care plan**
Goals, risks, supports, responsible person, follow-up interval and what to do if function suddenly changes.

- Scores are only useful when linked to a real decision: can the person live alone, bathe safely, manage medicines or attend appointments?
- Environmental fixes can compensate for impairment: grab bars, lighting, raised toilet seats, decluttering, alarms and transport support.
- Caregiver burden predicts whether a care plan is sustainable; assess the caregiver as part of the care system.
- Reassess after acute illness, hospitalization, falls, bereavement or medication changes.

Measure function > **Identify barriers** > **Match support** > **Monitor change**

doi:10.1093/geront/20.6.649; WHO ICOPE

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Quality of Life Assessment

From human outcomes to AI tasks

QoL is where ageing mechanisms, syndromes and function become visible as what matters to the person.

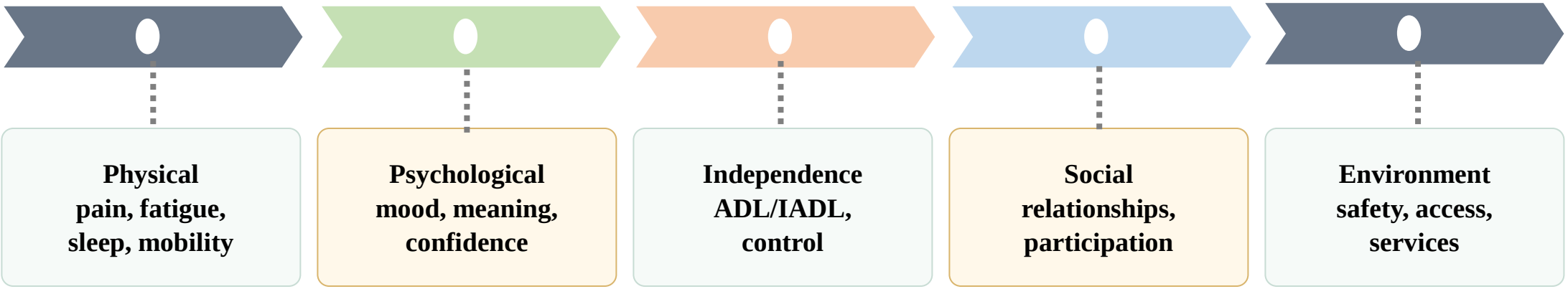
Function > **Experience** > **Goals** > **AI endpoint**

doi:10.1016/S0277-9536(98)00009-4

QoL is Broader Than Health, While HRQoL Focuses the Lens

Quality of Life (QoL)
A subjective, multidimensional judgement about how life is going in relation to goals, expectations, standards and concerns.

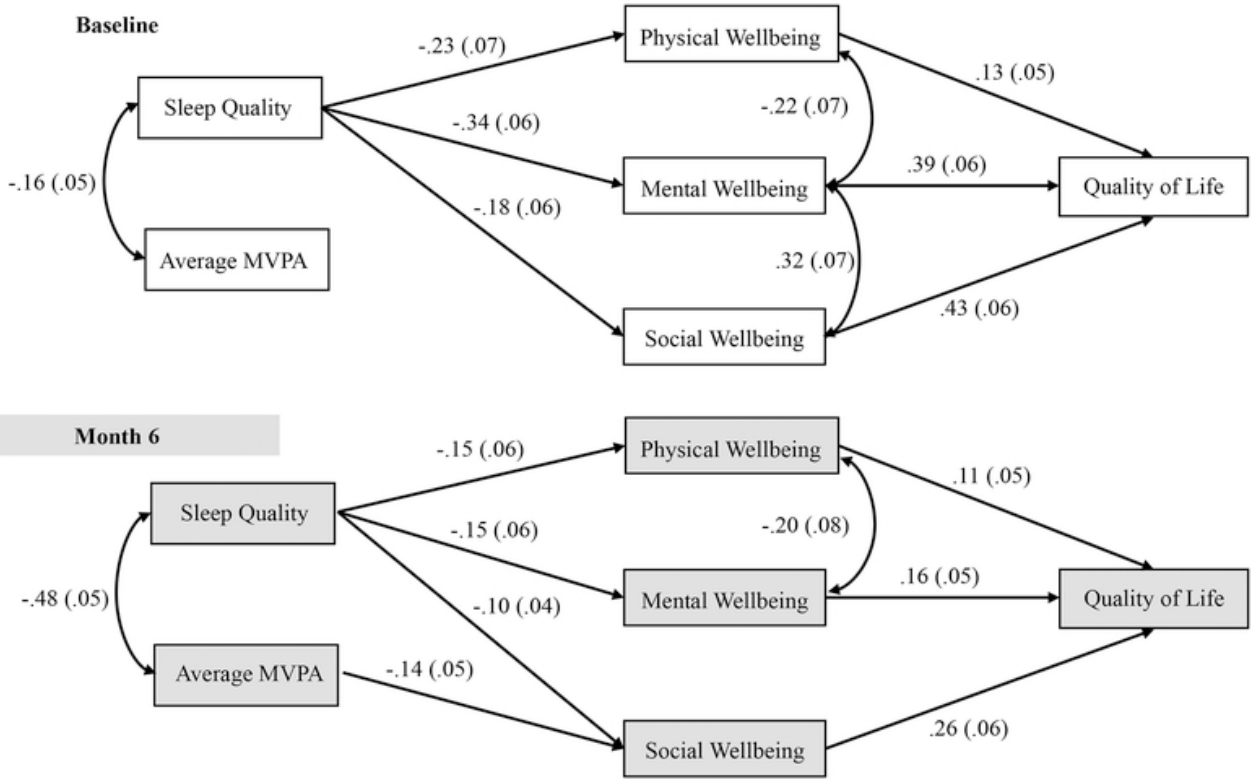
Health-Related QoL (HRQoL)
The part of QoL shaped by health status, symptoms, function, treatment burden and care context.



Teaching point: a treatment can improve disease markers yet fail the person if pain, sleep, independence or participation do not improve.

doi:10.1016/S0277-9536(98)00009-4

Pain, Sleep, Independence and Participation Shape QoL



Sleep and activity influence QoL through physical, mental and social well-being.

➢ **Why this matters in healthy ageing**

QoL is a distal outcome: it often changes through mediators such as pain control, sleep quality, autonomy, confidence and social connection.

- ➢ Pain affects movement, mood, sleep and willingness to participate.
- ➢ Poor sleep reduces energy, cognition and emotional regulation.
- ➢ Independence means control over daily routines, not only absence of disease.
- ➢ Participation asks whether the person can stay connected to valued roles.

Care question: which modifiable factor is most limiting the life this person wants to live?

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Macao Polytechnic University

doi:10.1080/15402002.2019.1690493

Common QoL Instruments Answer Different Questions

Instrument	Core Dimensions	What It Produces	Best Teaching Use
EQ-5D	Mobility, self-care, usual activities, pain, anxiety/depression	A health-state profile, VAS and utility value	Compact endpoint for cost-effectiveness and AI prediction
SF-36	Physical function, role, pain, general health, vitality, social, emotional, mental health	Eight scales plus physical and mental summary scores	Broader HRQoL profile for clinical studies
WHOQOL-OLD	Sensory abilities, autonomy, activities, social participation, death/dying, intimacy	Older-adult-specific QoL facet scores	Ageing-specific view of meaning, autonomy and participation

Rule of thumb: choose the instrument whose domains match the decision, population and intervention pathway.

doi:10.1016/0168-8510(90)90421-9; 10.1097/00005650-199206000-00002

QoL Is a Human-Centered Endpoint for Healthy Ageing AI

- **Predict**

  Who is likely to experience QoL decline after falls, insomnia, pain or functional loss?

- **Detect**

  Which sensor, EHR or speech patterns signal worsening sleep, participation or independence?

- **Recommend**

  Which intervention is most likely to improve the person's valued outcome?

- **Monitor**

  Is the care plan improving what the person reports, not only what the model measures?

Why QoL works as an endpoint?
It links clinical risk to lived benefit. A model that reduces falls but worsens confidence, sleep or autonomy may not improve healthy ageing.

- QoL labels are subjective and culturally shaped.
- PROMs may be missing when cognition, literacy or digital access is limited.
- Short-term optimization can conflict with long-term autonomy.
- Use QoL with function, safety and burden, not as a single magic label.

AI target: maximize meaningful function and experience, not just physiological normality.

doi:10.1016/S0277-9536(98)00009-4; PROMs

One Case Becomes Different Questions Across Disciplines

Discipline	How It Understands the Case	Variables or AI Task
Medicine	Falls, insomnia, polypharmacy, memory decline and comorbidity	Risk prediction, diagnosis support, medication review
Engineering	Mobility, sleep, home safety, wearable and environmental signals	Sensor fusion, anomaly detection, home monitoring
AI/Data Science	Labels, missingness, time windows, features and outcomes	Forecast decline, personalize interventions, evaluate benefit
Public Health	Access, caregiver capacity, social participation and equity	Population stratification, resource allocation, fairness

Translation habit: assessment item -> data variable -> clinical question -> AI task -> care action.

Section 1 Framework: From Mechanisms to Interventions

Ageing mechanisms
cellular senescence, inflammation, mitochondria

>

Geriatric syndromes
frailty, falls, delirium, sarcopenia, polypharmacy

>

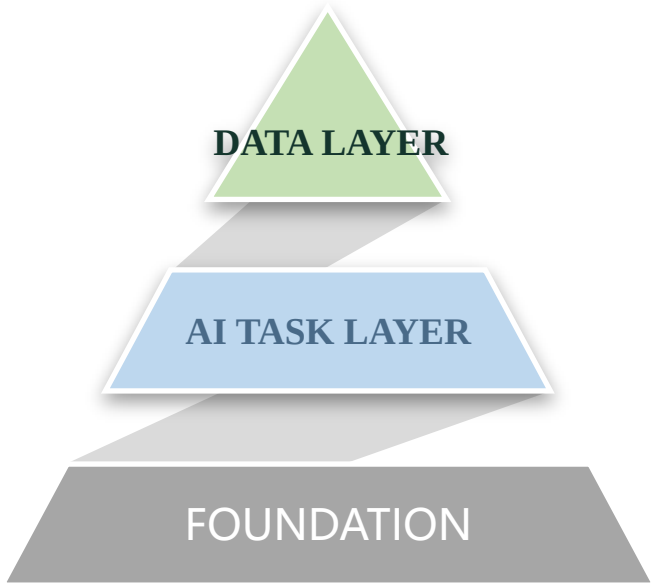
Functional status
ADL, IADL, mobility, cognition, support

>

Quality of life
pain, sleep, autonomy, participation

>

Intervention
care plan, rehabilitation, smart support



EHR + wearables + imaging + speech/behavior + scales + caregiver/environment data

classification, risk prediction, phenotyping, trajectory modeling, recommendation, monitoring

Transition to the next section: once the care questions are clear, data types become meaningful.

Framework synthesis: WHO healthy ageing + CGA + QoL/PROMs

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Thanks